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6

The gastrointestinal system

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Anatomy and physiology

The gastrointestinal system comprises the alimentary tract, liver, biliary system, pancreas and spleen. The alimentary tract extends from the mouth to the anus and includes the oesophagus, stomach, small intestine or small bowel (comprising the duodenum, jejunum and ileum), colon (large intestine or large bowel) and rectum (Figs 6.1–6.2 and Box 6.1).

The abdominal surface can be divided into nine regions by the intersection of two horizontal and two vertical planes (Fig. 6.1C).

The history

Gastrointestinal symptoms are common and are often caused by functional dyspepsia and irritable bowel syndrome. Alarm symptoms, indicating a more serious alternative or coexistent diagnosis, include persistent vomiting, dysphagia, gastrointestinal bleeding, weight loss, painless, watery, high-volume diarrhoea, nocturnal symptoms, fever and anaemia. The risk of serious disease increases with age. Always explore the patient's ideas, concerns and expectations about the symptoms (p. 5) to understand the clinical context.

Common presenting symptoms

Mouth symptoms

Bad breath (halitosis) due to gingival, dental or pharyngeal infection and dry mouth (xerostomia) are common mouth symptoms. Rarely patients complain of altered taste sensation (dysgeusia) or a foul taste in the mouth (cacogeusia).

Anorexia and weight loss

Anorexia is a loss of appetite and/or a lack of interest in food. In addition to enquiring about appetite, ask 'Do you still enjoy your food?'

Weight loss, in isolation, is rarely associated with serious organic disease. Ask how much weight has been lost, over what time. Loss of less than 3 kg in the previous 6 months is rarely significant. Weight loss is usually the result of reduced energy intake, not increased energy expenditure. It does not specifically indicate gastrointestinal disease, although it is common in many gastrointestinal disorders, including malignancy and liver disease. Energy requirements average 2500 kcal/day for males and 2000 kcal/day for females. Reduced energy intake arises from dieting, loss of appetite, malabsorption or malnutrition. Increased energy expenditure occurs in hyperthyroidism, fever or with the adoption of a more energetic lifestyle. A net calorie deficit of 1000 kcal/day results in weight loss of approximately 1 kg/week (7000 kcal $\cong$ 1 kg of fat). Greater weight loss during the initial stages of energy restriction arises from salt and water loss and depletion of hepatic glycogen stores, not from fat loss. Rapid weight loss over days suggests loss of body fluid as a result of vomiting, diarrhoea or diuretics (1 L of water = 1 kg). Check

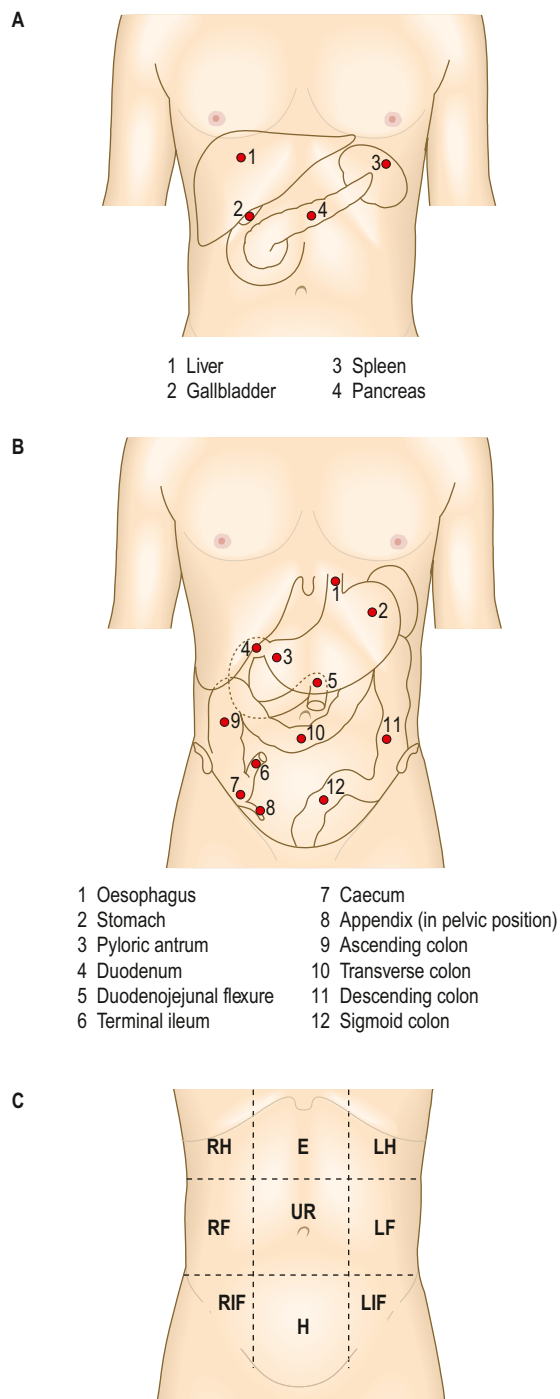


Fig. 6.1 Surface anatomy. **A** Abdominal surface markings of non-alimentary tract viscera. **B** Surface markings of the alimentary tract. **C** Regions of the abdomen. *E*, epigastrium; *H*, hypogastrium or suprapubic region; *LF*, left flank or lumbar region; *LH*, left hypochondrium; *LIF*, left iliac fossa; *RF*, right flank or lumbar region; *RH*, right hypochondrium; *RIF*, right iliac fossa; *UR*, umbilical region.

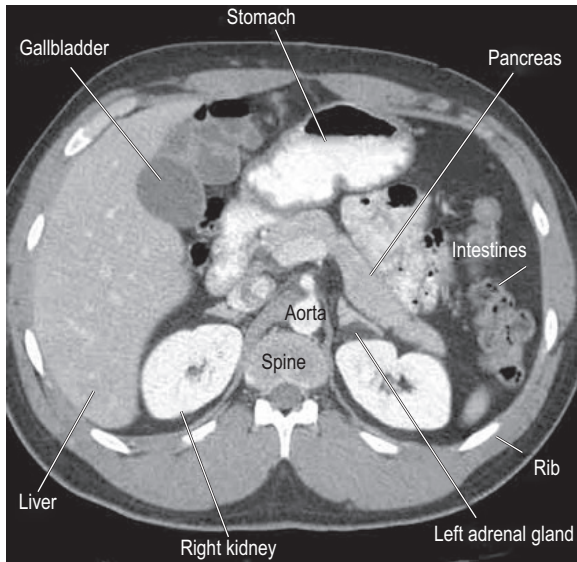


Fig. 6.2 Normal computed tomogram (CT) of the abdomen at L1 level.

6.1 Surface markings of the main non-alimentary tract abdominal organs

Structure	Position
Liver	Upper border: fifth right intercostal space on full expiration Lower border: at the costal margin in the mid-clavicular line on full inspiration
Spleen	Underlies left ribs 9–11, posterior to the mid-axillary line
Gallbladder	At the intersection of the right lateral vertical plane and the costal margin, i.e. tip of the ninth costal cartilage
Pancreas	The neck of the pancreas lies at the level of L1; the head lies below and right; the tail lies above and left
Kidneys	Upper pole lies deep to the 12th rib posteriorly, 7 cm from the midline; the right is 2–3 cm lower than the left

current and previous weight records to confirm apparent weight loss on examination (loose-fitting clothes, for example).

Pain

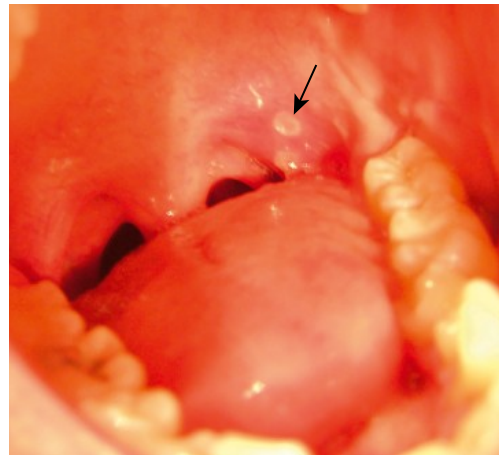
Painful mouth

Causes of sore lips, tongue or buccal mucosa include:

- deficiencies, including iron, folate, vitamin B₁₂ or C
- dermatological disorders, including lichen planus (Fig. 6.3A)
- chemotherapy
- aphthous ulcers (Fig. 6.3B)
- infective stomatitis
- inflammatory bowel disease and coeliac disease, associated with mouth ulcers.



A



B

Fig. 6.3 Some causes of a painful mouth. **A** Lichen planus. **B** Small, 'punched-out' aphthous ulcer (arrow).

Heartburn and reflux

Heartburn is a hot, burning retrosternal discomfort.

To differentiate heartburn from cardiac chest pain, ask about associated features:

- character of pain: burning
- radiation: upward
- precipitating factors: lying flat or bending forward
- associated symptoms:
 - waterbrash (sudden appearance of fluid in the mouth due to reflex salivation as a result of gastro-oesophageal reflux disease (GORD) or, rarely, peptic ulcer disease)
 - the taste of acid appearing in the mouth due to reflux/regurgitation.

When heartburn is the principal symptom, GORD is the most likely diagnosis.

Dyspepsia

Dyspepsia is pain or discomfort centred in the upper abdomen. In contrast, 'indigestion' is a term commonly used by patients for ill-defined symptoms from the upper gastrointestinal tract.

Ask about:

- site of pain
- character of pain
- exacerbating and relieving factors, such as food and antacid
- associated symptoms, such as nausea, belching, bloating and premature fullness (early satiety).

Clusters of symptoms are used to classify dyspepsia:

- reflux-like dyspepsia (heartburn-predominant dyspepsia)
- ulcer-like dyspepsia (epigastric pain relieved by food or antacids)
- dysmotility-like dyspepsia (nausea, belching, bloating and premature fullness (early satiety).

Often there is no structural cause and the dyspepsia is functional. Patients below the age of 55 without alarm symptoms and with a negative *Helicobacter pylori* test can be positively diagnosed as having functional dyspepsia thus avoiding unnecessary investigations but if symptoms persist then further investigations should be considered. However, in patients over the age of 55 organic pathology should always be excluded by upper gastrointestinal (GI) endoscopy.

Dyspepsia that is worse with an empty stomach and eased by eating is typical of peptic ulceration. The patient may indicate a single localised point in the epigastrium (pointing sign) and complain of nausea and abdominal fullness that is worse after fatty or spicy meals. 'Fat intolerance' is common with all causes of dyspepsia, including gallbladder disease.

Odynophagia

Odynophagia is pain from swallowing, often precipitated by drinking hot liquids. It can be present with or without dysphagia (see below) and may indicate oesophageal ulceration or oesophagitis from gastro-oesophageal reflux or oesophageal

candidiasis. It implies intact mucosal sensation, making oesophageal cancer unlikely.

Abdominal pain

Characterise the pain using the acronym SOCRATES (see Box 2.2). Ask about the characteristics described here.

Site

Visceral abdominal pain from distension of hollow organs, mesenteric traction or excessive smooth-muscle contraction is deep and poorly localised in the midline. The pain is conducted via sympathetic splanchnic nerves. Somatic pain from the parietal peritoneum and abdominal wall is lateralised and localised to the inflamed area. It is conducted via intercostal nerves.

Pain arising from foregut structures (stomach, pancreas, liver and biliary system) is localised above the umbilicus (Fig. 6.4). Central abdominal pain arises from midgut structures, such as the small bowel and appendix. Lower abdominal pain arises from hindgut structures, such as the colon. Inflammation may cause localised pain: for example, left iliac fossa pain due to diverticular disease of the sigmoid colon.

Pain from an unpaired structure, such as the pancreas, is midline and radiates through to the back. Pain from paired structures, such as renal colic, is felt on, and radiates to, the affected side (Fig. 6.5). Torsion of the testis may present with abdominal pain (p. 267). In females, consider gynaecological causes such as ruptured ovarian cyst, pelvic inflammatory disease, endometriosis or ectopic pregnancy (p. 247).

Onset

Sudden onset of severe abdominal pain, rapidly progressing to become generalised and constant, suggests a hollow viscus perforation (usually due to peptic ulceration, diverticular disease or colorectal cancer), a ruptured abdominal aortic aneurysm or mesenteric infarction.

Torsion of the caecum or sigmoid colon (volvulus) presents with sudden abdominal pain associated with acute intestinal obstruction.

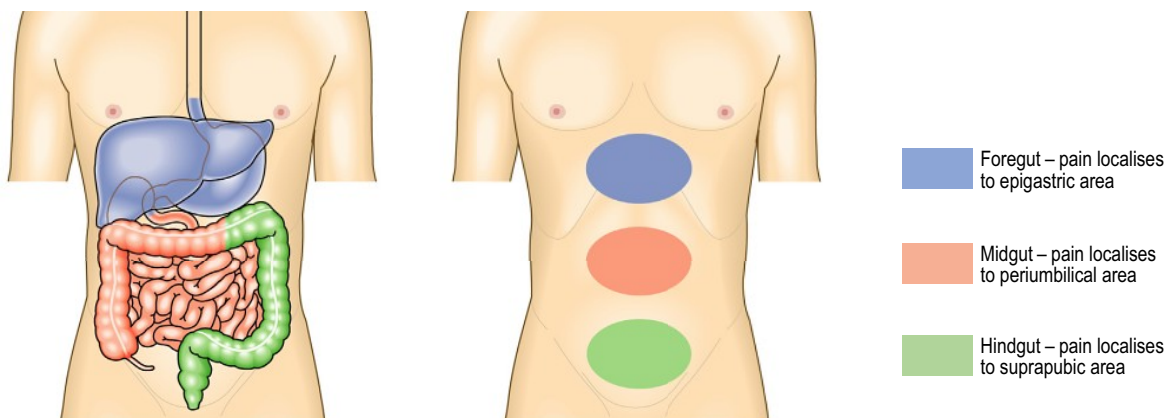


Fig. 6.4 Abdominal pain. Perception of visceral pain is localised to the epigastric, umbilical or suprapubic region, according to the embryological origin of the affected organ.

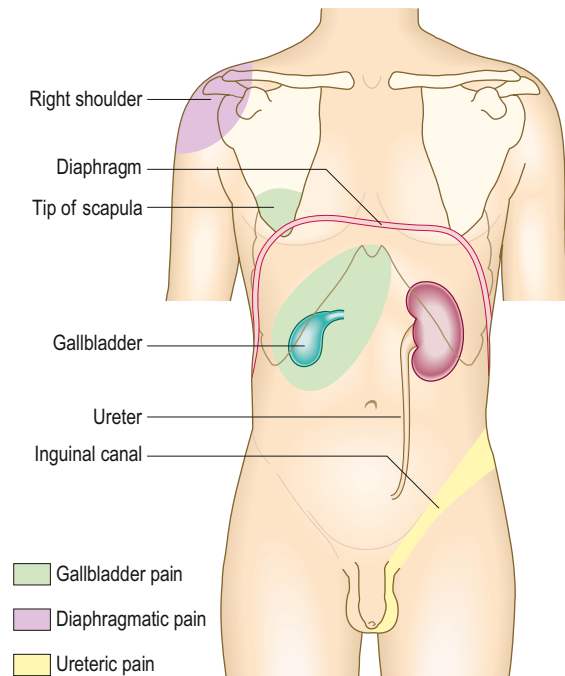


Fig. 6.5 Characteristic radiation of pain from the gallbladder, diaphragm and ureters.

Character

Colicky pain lasts for a short time (seconds or minutes), eases off and then returns. It arises from hollow structures, as in small or large bowel obstruction, or the uterus during labour.

Biliary and renal 'colic' are misnamed, as the pain is rarely colicky; pain rapidly increases to a peak and persists over several hours before gradually resolving. Dull, constant, vague and poorly localised pain is more typical of an inflammatory process or infection, such as pelvic inflammatory disease, appendicitis or diverticulitis (Box 6.2).

Radiation

Pain radiating from the right hypochondrium to the shoulder or interscapular region may reflect diaphragmatic irritation, as in acute cholecystitis (Fig. 6.5). Pain radiating from the loin to the groin and genitalia is typical of renal colic. Central upper abdominal pain radiating through to the back, partially relieved by sitting forward, suggests pancreatitis. Central abdominal pain that later shifts into the right iliac fossa occurs in acute appendicitis. The combination of severe back and abdominal pain may indicate a ruptured or dissecting abdominal aortic aneurysm.

Associated symptoms

Anorexia, nausea and vomiting are common but non-specific symptoms. They may accompany any very severe pain but conversely may be absent, even in advanced intra-abdominal disease. Abdominal pain due to irritable bowel syndrome,

6.2 Diagnosing abdominal pain

Disorder				
	Peptic ulcer	Biliary colic	Acute pancreatitis	Renal colic
Site	Epigastrium	Epigastrium/right hypochondrium	Epigastrium/left hypochondrium	Loin
Onset	Gradual	Rapidly increasing	Sudden	Rapidly increasing
Character	Gnawing	Constant	Constant	Constant
Radiation	Into back	Below right scapula	Into back	Into genitalia and inner thigh
Associated symptoms	Non-specific	Non-specific	Non-specific	Non-specific
Timing				
Frequency/periodicity	Remission for weeks/months	Attacks can be enumerated	Attacks can be enumerated	Usually a discrete episode
Special times	Nocturnal and especially when hungry	Unpredictable	After heavy drinking	Following periods of dehydration
Duration	1/2–2 hours	4–24 hours	>24 hours	4–24 hours
Exacerbating factors	Stress, spicy foods, alcohol, non-steroidal anti-inflammatory drugs	Eating – unable to eat during bouts	Alcohol Eating – unable to eat during bouts	–
Relieving factors	Food, antacids, vomiting	–	Sitting upright	–
Severity	Mild to moderate	Severe	Severe	Severe

diverticular disease or colorectal cancer is usually accompanied by altered bowel habit. Other features such as breathlessness or palpitation suggest non-alimentary causes (Box 6.3).

Hypotension and tachycardia following the onset of pain suggest intra-abdominal sepsis or bleeding: for example, from a peptic ulcer, a ruptured aortic aneurysm or an ectopic pregnancy.

Timing

During the first 1–2 hours after perforation, a 'silent interval' may occur when abdominal pain resolves transiently. The initial chemical peritonitis may subside before bacterial peritonitis becomes established. For example, in acute appendicitis, pain is initially periumbilical (visceral pain) and moves to the right iliac fossa (somatic pain) when localised inflammation of the parietal peritoneum becomes established. If the appendix ruptures, generalised peritonitis may develop. Occasionally, a localised appendix abscess develops, with a palpable mass and localised pain in the right iliac fossa.

6.3 Non-alimentary causes of abdominal pain

Disorder	Clinical features
Myocardial infarction	Epigastric pain without tenderness <i>Angor animi</i> (feeling of impending death) Hypotension Cardiac arrhythmias
Dissecting aortic aneurysm	Tearing interscapular pain <i>Angor animi</i> Hypotension Asymmetry of femoral pulses
Acute vertebral collapse	Lateralised pain restricting movement Tenderness overlying involved vertebra
Cord compression	Pain on percussion of thoracic spine Hyperaesthesia at affected dermatome with sensory loss below Spinal cord signs
Pleurisy	Lateralised pain on coughing Chest signs (e.g. pleural rub)
Herpes zoster	Hyperaesthesia in dermatomal distribution Vesicular eruption
Diabetic ketoacidosis	Cramp-like pain Vomiting Air hunger Tachycardia Ketotic breath
Pelvic inflammatory disease or tubal pregnancy	Suprapubic and iliac fossa pain, localised tenderness Nausea, vomiting Fever
Torsion of testis/ovary	Lower abdominal pain Nausea, vomiting Localised tenderness

A change in the pattern of symptoms suggests either that the initial diagnosis was wrong or that complications have developed. In acute small bowel obstruction, a change from typical intestinal colic to persistent pain with abdominal tenderness suggests intestinal ischaemia, as in strangulated hernia, and is an indication for urgent surgical intervention.

Abdominal pain persisting for hours or days suggests an inflammatory disorder, such as acute appendicitis, cholecystitis or diverticulitis.

Exacerbating and relieving factors

Pain exacerbated by movement or coughing suggests inflammation. Patients tend to lie still to avoid exacerbating the pain. People with colic typically move around or draw their knees up towards the chest during spasms.

Severity

Excruciating pain, poorly relieved by opioid analgesia, suggests an ischaemic vascular event, such as bowel infarction or ruptured abdominal aortic aneurysm. Severe pain rapidly eased by potent analgesia is more typical of acute pancreatitis or peritonitis secondary to a ruptured viscus.

Features of the pain can help distinguish between possible causes (Box 6.3).

The acute abdomen

The majority of general surgical emergencies are patients with sudden severe abdominal pain (an 'acute abdomen'). Patients may be so occupied by recent and severe symptoms that they forget important details of their history unless asked directly. Seek additional information from family or friends if severe pain, shock or altered consciousness makes it difficult to obtain a history from the patient. Note any relevant past history, such as known diverticular disease in a patient with a possible acute perforation. Causes range from self-limiting to severe life-threatening diseases (Box 6.4). Evaluate patients rapidly, and then resuscitate critically ill patients immediately before undertaking further assessment and surgical intervention. Parenteral opioid analgesia to alleviate severe abdominal pain will help, not hinder, clinical assessment. In patients with undiagnosed acute abdominal pain, reassess their clinical state regularly, undertake urgent investigations and consider surgical intervention in a timely fashion.

Dysphagia

Patients with dysphagia complain that food or drink sticks when they swallow.

Ask about:

- onset: recent or longstanding
- nature: intermittent or progressive
- difficulty swallowing solids, liquids or both
- the level where food is felt to stick
- any regurgitation or reflux of food or fluid
- any associated pain (odynophagia) or heartburn
- any recent weight loss.
- past history of food bolus obstruction

6.4 Typical clinical features in patients with an 'acute abdomen'

Condition	History	Examination
Acute appendicitis	Nausea, vomiting, central abdominal pain that later shifts to the right iliac fossa	Fever, tenderness, guarding or palpable mass in the right iliac fossa, pelvic peritonitis on rectal examination
Perforated peptic ulcer with acute peritonitis	Vomiting at onset associated with severe acute-onset abdominal pain, previous history of dyspepsia, ulcer disease, non-steroidal anti-inflammatory drugs or glucocorticoid therapy	Shallow breathing with minimal abdominal wall movement, abdominal tenderness and guarding, board-like rigidity, abdominal distension and absent bowel sounds
Acute pancreatitis	Anorexia, nausea, vomiting, constant severe epigastric pain, previous alcohol abuse/cholelithiasis	Fever, periumbilical or loin bruising, epigastric tenderness, variable guarding, reduced or absent bowel sounds
Ruptured aortic aneurysm	Sudden onset of severe, tearing back/loin/abdominal pain, hypotension and past history of vascular disease and/or high blood pressure	Shock and hypotension, pulsatile, tender, abdominal mass, asymmetrical femoral pulses
Acute mesenteric ischaemia	Anorexia, nausea, vomiting, bloody diarrhoea, constant abdominal pain, previous history of vascular disease and/or high blood pressure	Atrial fibrillation, heart failure, asymmetrical peripheral pulses, absent bowel sounds, variable tenderness and guarding
Intestinal obstruction	Colicky central abdominal pain, nausea, vomiting and constipation	Surgical scars, hernias, mass, distension, visible peristalsis, increased bowel sounds
Ruptured ectopic pregnancy	Premenopausal female, delayed or missed menstrual period, hypotension, unilateral iliac fossa pain, pleuritic shoulder-tip pain, 'prune juice'-like vaginal discharge	Suprapubic tenderness, periumbilical bruising, pain and tenderness on vaginal examination (cervical excitation), swelling/fullness in fornix on vaginal examination
Pelvic inflammatory disease	Sexually active young female, previous history of sexually transmitted infection, recent gynaecological procedure, pregnancy or use of intrauterine contraceptive device, irregular menstruation, dyspareunia, lower or central abdominal pain, backache, pleuritic right upper quadrant pain (Fitz-Hugh–Curtis syndrome)	Fever, vaginal discharge, pelvic peritonitis causing tenderness on rectal examination, right upper quadrant tenderness (perihepatitis), pain/tenderness on vaginal examination (cervical excitation), swelling/fullness in fornix on vaginal examination

Do not confuse dysphagia with early satiety, the inability to complete a full meal because of premature fullness, or with globus, which is a feeling of a lump in the throat. Globus does not interfere with swallowing and is not related to eating.

Neurological dysphagia resulting from bulbar or pseudobulbar palsy (p. 140) is worse for liquids than solids and may be accompanied by choking, spluttering and fluid regurgitating from the nose.

Neuromuscular dysphagia, or oesophageal dysmotility, presents in middle age, is worse for solids and may be helped by liquids and sitting upright. Achalasia, when the lower oesophageal sphincter fails to relax normally, typically results in dysphagia for both solids and liquids and leads to progressive oesophageal dilatation above the sphincter. Overflow of secretions and food into the respiratory tract may then occur, especially at night when the patient lies down, causing aspiration pneumonia. Oesophageal dysmotility and acid reflux can provoke oesophageal spasms and central chest pain, which may be confused with cardiac pain.

A pharyngeal pouch may cause food to stick or be regurgitated undigested from previous days and may lead to recurrent chest infections due to chronic silent aspiration.

'Mechanical' dysphagia is often due to oesophageal stricture but can be caused by external compression. With weight loss, a short history and no reflux symptoms, suspect oesophageal

cancer. Longstanding dysphagia without weight loss but accompanied by heartburn is more likely to be due to benign peptic stricture. Eosinophilic oesophagitis is the most common cause of food bolus obstruction and should be considered in younger patients with dysphagia; it is associated with atopy and food allergy. Record the site at which the patient feels the food sticking; although this is not an entirely reliable guide to the site of obstruction. If dysphagia is experienced high in the neck, consider tumours of the pharynx or larynx or extrinsic compression from a mass lesion such as a thyroid goitre.

Nausea and vomiting

Nausea is the sensation of feeling sick. Vomiting is the expulsion of gastric contents via the mouth. Both are associated with pallor, sweating and hyperventilation.

Ask about:

- relation to meals and timing, such as early morning or late evening
- associated symptoms, such as dyspepsia and abdominal pain, and whether they are relieved by vomiting
- whether the vomit is bile-stained (green), blood-stained or faeculent
- associated weight loss
- the patient's medications.

Nausea and vomiting, particularly with abdominal pain or discomfort, suggest upper gastrointestinal disorders. Dyspepsia causes nausea without vomiting. Peptic ulcers seldom cause painless vomiting unless they are complicated by pyloric stenosis, which causes projectile vomiting of large volumes of gastric content that is not bile-stained. Obstruction distal to the pylorus produces bile-stained vomit. Severe vomiting without significant pain suggests gastric outlet or proximal small bowel obstruction. Faeculent vomiting of small bowel contents (not faeces) is a late feature of distal small bowel or colonic obstruction. In peritonitis, the vomitus is usually small in volume but persistent. The more distal the level of intestinal obstruction, the more marked the accompanying abdominal distension and colic.

Vomiting is common in gastroenteritis, cholecystitis, pancreatitis and hepatitis. It is typically preceded by nausea but raised intracranial pressure may occur without warning. Severe pain may precipitate vomiting, as in renal or biliary colic or myocardial infarction.

Anorexia nervosa and bulimia are eating disorders characterised by undisclosed, self-induced vomiting. In bulimia, weight is maintained or increased, unlike in anorexia nervosa, where profound weight loss is common.

Other non-gastrointestinal causes of nausea and vomiting include:

- drugs, such as opioids, theophyllines, digoxin, cytotoxic agents, antidepressants or alcohol
- pregnancy
- diabetic ketoacidosis
- renal or liver failure
- hypercalcaemia
- Addison's disease
- raised intracranial pressure (meningitis, brain tumour)
- vestibular disorders (labyrinthitis and Ménière's disease).

Wind and flatulence

Belching, excessive or offensive flatus, abdominal distension and borborygmi (audible bowel sounds) are often called 'wind' or flatulence. Clarify exactly what the patient means. Belching is due to air swallowing (aerophagy) and has no medical significance. It may indicate anxiety but sometimes occurs in an attempt to relieve abdominal pain or discomfort and accompanies GORD.

Normally, 200–2000 mL of flatus is passed each day. Flatus is a mixture of gases derived from swallowed air and colonic bacterial fermentation of poorly absorbed carbohydrates. Excessive flatus occurs particularly in lactase deficiency and intestinal malabsorption.

Borborygmi result from the movement of fluid and gas along the bowel. Loud borborygmi, particularly if associated with colicky discomfort, suggest small bowel obstruction or dysmotility.

Abdominal distension

Abdominal girth slowly increasing over months or years is usually due to obesity but in a patient with weight loss, it suggests intra-

abdominal disease. The most common causes of abdominal distension are:

- fat due to obesity
- flatus due to pseudo-obstruction or bowel obstruction
- faeces due to subacute obstruction or constipation
- fluid due to ascites (accumulation of fluid in the peritoneal cavity; Fig. 6.6), tumours (especially ovarian) or a distended bladder
- fetus
- functional bloating (fluctuating abdominal distension that develops during the day and resolves overnight, usually occurring in irritable bowel syndrome).

Altered bowel habit

Diarrhoea

Clarify what patients mean by diarrhoea. They may complain of frequent stools or a change in the consistency of the stools. Normal frequency ranges from three bowel movements daily to once every 3 days. Diarrhoea is the frequent passage of loose stools. Steatorrhoea is diarrhoea associated with fat malabsorption. The stools are greasy, pale and bulky, and they float, making them difficult to flush away.

Ask about:

- onset of diarrhoea: acute, chronic or intermittent
- stool:
 - frequency
 - volume
 - colour
 - consistency: watery, unformed or semisolid
 - contents: red blood, mucus or pus

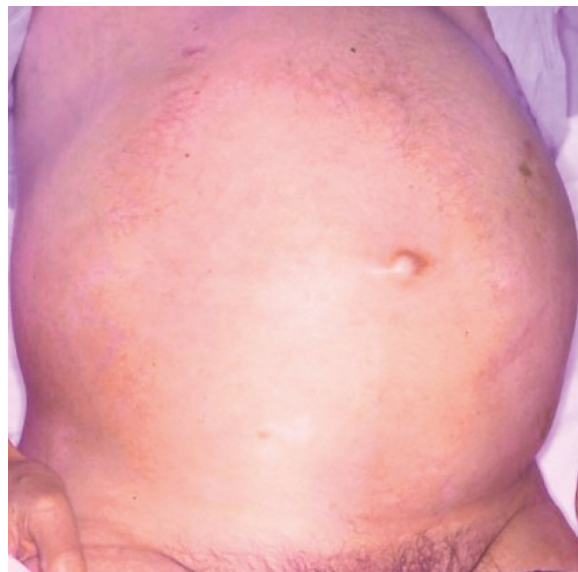


Fig. 6.6 Abdominal distension due to ascites.

- associated features: urgency, faecal incontinence or tenesmus (the sensation of needing to defecate, although the rectum is empty), abdominal pain, vomiting, sleep disturbance
- recent travel and where to
- recent medication, in particular any antibiotics.

High-volume diarrhoea (>1 L/day) occurs when the stool water content is increased (the principal site of physiological water absorption being the colon) and may be:

- secretory, due to intestinal inflammation, as in infection or inflammatory bowel disease
- osmotic, due to malabsorption, drugs (as in laxative abuse) or motility disorders (autonomic neuropathy, particularly in diabetes).

If the patient fasts, osmotic diarrhoea stops but secretory diarrhoea persists. The most common cause of acute diarrhoea is infective gastroenteritis due to norovirus, *Salmonella* species or *Clostridium difficile*. Infective diarrhoea can become chronic (>4 weeks) in cases of parasitic infestations (such as giardiasis (*Giardia lamblia*), amoebiasis or cryptosporidiosis). Steatorrhoea is common in coeliac disease, chronic pancreatitis and pancreatic insufficiency due to cystic fibrosis. Bloody diarrhoea may be caused by inflammatory bowel disease, colonic ischaemia or infective gastroenteritis. Change in the bowel habit towards diarrhoea can be a manifestation of colon cancer, in particular, cancer of the right side of the colon and in patients over 50 years. Thyrotoxicosis is often accompanied by secretory diarrhoea or steatorrhoea and weight loss.

Low-volume diarrhoea is associated with irritable bowel syndrome. Abdominal pain, bloating, dyspepsia and non-alimentary symptoms commonly accompany irritable bowel symptoms. Criteria have been developed to define irritable bowel syndrome more precisely, taking into account the duration of symptoms, the presence of abdominal pain and its relationship to defecation, and the frequency and consistency of stools (see Rome IV criteria for irritable bowel syndrome).

Constipation

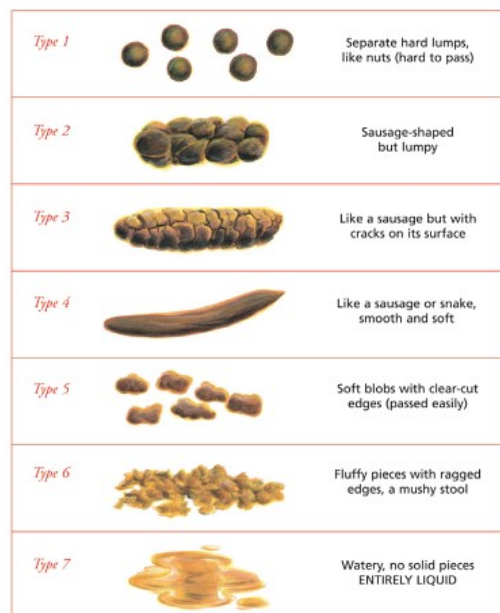
Clarify what the patient means by constipation. Use the Bristol stool form scale (Fig. 6.7) to describe the stools. Constipation is the infrequent passage of hard stools.

Ask about:

- onset: lifelong or of recent onset
- stool frequency: how often the patient moves their bowels each week and how much time is spent straining at stool
- shape of the stool: for example, pellet-like
- associated symptoms, such as abdominal pain, anal pain on defecation or rectal bleeding
- drugs that may cause constipation.

Constipation may be due to lack of dietary fibre, impaired colonic motility, mechanical intestinal obstruction, impaired rectal sensation or anorectal dysfunction impairing the process of defecation. Constipation is common in irritable bowel syndrome. Other important causes include colorectal cancer, hypothyroidism, hypercalcaemia, drugs (opiates, iron) and immobility (Parkinson's disease, stroke). Absolute constipation (no flatus or

THE BRISTOL STOOL FORM SCALE



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Fig. 6.7 Bristol stool form scale. Reproduced with kind permission of Dr KW Heaton, formerly Reader in Medicine at the University of Bristol. ©2000, Norgine group of companies.

bowel movements) suggests intestinal obstruction and is usually associated with pain, vomiting and distension. Tenesmus suggests rectal inflammation or tumour. Faecal impaction can occasionally present as overflow diarrhoea.

Bleeding

Haematemesis

Haematemesis is the vomiting of blood.

Ask about:

- Colour: is the vomitus fresh red blood or dark brown, resembling coffee grounds?
- Onset: was haematemesis preceded by intense retching or was blood staining apparent in the first vomit?
- History of dyspepsia, peptic ulceration, gastrointestinal bleeding or liver disease.
- Alcohol, non-steroidal anti-inflammatory drugs (NSAIDs) and glucocorticoid ingestion.

If the source of bleeding is above the gastro-oesophageal sphincter, as with oesophageal varices, fresh blood may well up in the mouth, as well as being actively vomited. With a lower

oesophageal mucosal tear due to the trauma of forceful retching (Mallory–Weiss syndrome), fresh blood appears only after the patient has vomited forcefully several times.

Melaena

Melaena is the passage of tarry, shiny black stools with a characteristic odour and results from upper gastrointestinal bleeding. Distinguish this from the matt black stools associated with oral iron or bismuth therapy.

Peptic ulceration (gastric or duodenal) is the most common cause of upper gastrointestinal bleeding and can manifest with melaena, haematemesis or both. Excessive alcohol ingestion may cause haematemesis from erosive gastritis, Mallory–Weiss tear or bleeding oesophagogastric varices in cirrhotic patients. Oesophageal or gastric cancer, gastric angiodectasias (Dieulafoy lesion when localised in the fundus) and portal hypertensive

gastropathy due to splenic vein thrombosis are rare causes of upper gastrointestinal bleeding.

The Rockall and Blatchford scores are used to assess the risk of gastrointestinal bleeding (Boxes 6.5 and 6.6). A profound upper gastrointestinal bleed may lead to the passage of purple stool or, rarely, fresh blood.

Rectal bleeding

Establish whether the blood is mixed with stool, coats the surface of the otherwise normal stool or is seen on the toilet paper or in the pan. Fresh rectal bleeding (haematochezia) usually indicates a disorder in the anal canal, rectum or colon. During severe upper gastrointestinal bleeding, however, blood may pass through the intestine unaltered, causing fresh rectal

6.5 Prediction of the risk of mortality in patients with upper gastrointestinal bleeding: Rockall score

Criterion	Score
Age	
<60 years	0
60–79 years	1
>80 years	2
Shock	
None	0
Pulse >100 beats per minute and systolic blood pressure >100 mmHg	1
Systolic blood pressure <100 mmHg	2
Comorbidity	
None	0
Heart failure, ischaemic heart disease or other major illness	2
Renal failure or disseminated malignancy	3
Endoscopic findings	
Mallory–Weiss tear and no visible bleeding	0
All other diagnoses	1
Upper gastrointestinal malignancy	2
Major stigmata of recent haemorrhage	
None	0
Visible bleeding vessel/adherent clot	2
Total score	
Pre-endoscopy (maximum score = 7)	Score 4 = 14% mortality pre-endoscopy
Post-endoscopy (maximum score = 11)	Score 8+ = 25% mortality post-endoscopy
Reproduced from Rockall TA, Logan RF, Devlin HB, et al. Risk assessment after acute upper gastrointestinal haemorrhage. <i>Journal of the British Society of Gastroenterology</i> 1996; 38(3):316, with permission from BMJ Publishing Group Ltd.	

6.6 Prediction of need to have a medical intervention (e.g. blood transfusion or endoscopy) in patients with upper gastrointestinal bleeding: The Glasgow–Blatchford bleeding score

Admission risk marker	Score component value
Blood urea (mmol/L)	
>6–5 <8–0	2
>8–0 <10–0	3
<1–0 <25–0	4
>25	6
Haemoglobin (g/L) for men	
>120 <130	1
>100 <120	3
<10–0	6
Haemoglobin (g/L) for women	
≥100<120	1
<100	6
Systolic blood pressure (mmHg)	
<100–109	1
90–99	2
<90	3
Other markers	
Pulse >100 (per min)	1
Presentation with melaena	1
Presentation with syncope	2
Hepatic disease	2
Cardiac failure	2
Score of 0 no interventions needed; patient can be treated as an outpatient	
Scores of 6 >50% risk of needing an intervention.	
Reprinted with permission from Elsevier, Blatchford O, Murray WR, Blatchford M. A risk score to predict need for treatment for upper-gastrointestinal haemorrhage. <i>Lancet</i> 2000;356:1318–21.	

bleeding. Common causes of rectal bleeding include haemorrhoids, anal fissures (blood on the toilet paper or in the pan), complicated diverticular disease, colorectal cancer or colonic polyps, inflammatory bowel disease, ischaemic colitis and colonic angioectasias.

Jaundice

Jaundice is a yellowish discolouration of the skin, sclerae (Fig. 6.8) and mucous membranes caused by hyperbilirubinaemia (Box 6.7). There is no absolute level at which jaundice is clinically detected but, in good light, most clinicians will recognise jaundice when bilirubin levels exceed 50 $\mu\text{mol/L}$ (2.92 mg/dL).

Ask about:

- associated symptoms: abdominal pain, fever, weight loss, itching
- colour of stools (normal or pale) and urine (normal or dark)
- alcohol intake
- travel history and immunisations
- use of illicit or intravenous drugs
- sexual history
- previous blood transfusions
- recently prescribed drugs.

Unconjugated bilirubin is insoluble and binds to plasma albumin; it is therefore not filtered by the renal glomeruli. In jaundice from unconjugated hyperbilirubinaemia, the urine is a normal colour (acholuric jaundice; Box 6.8).

Bilirubin is conjugated to form bilirubin diglucuronide in the liver and excreted in bile, producing its characteristic green colour. In conjugated hyperbilirubinaemia, the urine is dark brown due to the presence of bilirubin diglucuronide. In the colon, conjugated bilirubin is metabolised by bacteria to stercobilinogen and stercobilin, which contribute to the brown colour of stool. Stercobilinogen is absorbed from the bowel and excreted in the urine as urobilinogen, a colourless, water-soluble compound.

Prehepatic jaundice

In haemolytic disorders, anaemic pallor combined with jaundice may produce a pale lemon complexion. The stools and urine are

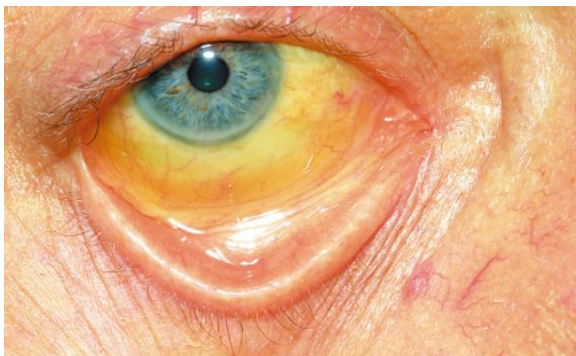


Fig. 6.8 Yellow sclera of jaundice.

6.7 Common causes of jaundice

Increased bilirubin production

- Haemolysis (unconjugated hyperbilirubinaemia)

Impaired bilirubin excretion

- Congenital:
 - Gilbert's syndrome (unconjugated)
- Hepatocellular:
 - Viral hepatitis
 - Cirrhosis
 - Drugs
 - Autoimmune hepatitis
- Intrahepatic cholestasis:
 - Drugs
 - Primary biliary cirrhosis
- Extrahepatic cholestasis:
 - Gallstones
 - Cancer: pancreas, cholangiocarcinoma

normal in colour. Gilbert's syndrome is common and causes unconjugated hyperbilirubinaemia. Serum liver enzyme concentrations are normal and jaundice is mild (plasma bilirubin <100 $\mu\text{mol/L}$ (5.85 mg/dL)) but increases during prolonged fasting or intercurrent febrile illness.

Hepatic jaundice

Hepatocellular disease causes hyperbilirubinaemia that is both unconjugated and conjugated. Conjugated bilirubin leads to dark brown urine. The stools are normal in colour.

Posthepatic/cholestatic jaundice

In biliary obstruction, conjugated bilirubin in the bile does not reach the intestine, so the stools are pale. Obstructive jaundice may be accompanied by pruritus (generalised itch) due to skin deposition of bile salts. Obstructive jaundice with abdominal pain is usually due to gallstones; if fever or rigours also occur (Charcot's triad), ascending cholangitis is likely. Painless obstructive jaundice suggests malignant biliary obstruction, as in cholangiocarcinoma or cancer of the head of the pancreas. Obstructive jaundice can be due to intrahepatic as well as extrahepatic cholestasis, as in primary biliary cirrhosis, certain hepatotoxic drug reactions (Box 6.9) and profound hepatocellular injury.

Groin swellings and lumps

Ask about:

- associated pain

6.8 Urine and stool analysis in jaundice

	Urine			Stools
	Colour	Bilirubin	Urobilinogen	Colour
Unconjugated	Normal	–	++++	Normal
Hepatocellular	Dark	++	++	Normal
Obstructive	Dark	++++	–	Pale

6.9 Examples of drug-induced gastrointestinal conditions

Symptom	Drug
Weight gain	Oral glucocorticoids
Dyspepsia and gastrointestinal bleeding	Aspirin Non-steroidal anti-inflammatory drugs
Nausea	Many drugs, including selective serotonin reuptake inhibitor antidepressants
Diarrhoea (pseudomembranous colitis)	Antibiotics Proton pump inhibitors
Constipation	Opioids
Jaundice: hepatitis	Paracetamol (overdose) Pyrazinamide Rifampicin Isoniazid
Jaundice: cholestatic	Flucloxacillin Chlorpromazine Co-amoxiclav
Liver fibrosis	Methotrexate

- precipitating/exacerbating factors, such as straining due to chronic constipation, chronic cough, heavy manual labour and relationship with micturition
- timing: when the symptoms are worse

Hernias are common causes of groin lumps and frequently present with dull, dragging discomfort (rather than acute pain), which is often exacerbated by straining and after long periods of standing or activity. Patients can often manually reduce the hernia by applying gentle pressure over the swelling or by lying flat. Other causes of groin swellings include lymph nodes, skin and subcutaneous lumps and, less commonly, saphena varix (a varicosity of the long saphenous vein), hydrocoele of the spermatic cord, undescended testis, femoral aneurysm or a psoas abscess.

Past medical history

History of a similar problem may suggest the diagnosis: for example, pancreatitis, bleeding peptic ulcer or inflammatory bowel disease. Coexisting peripheral vascular disease, hypertension, heart failure or atrial fibrillation may suggest aortic aneurysm or mesenteric ischaemia as the cause of acute abdominal pain. Primary biliary cirrhosis and autoimmune hepatitis are associated with thyroid disease, and non-alcoholic fatty liver disease (NAFLD) is associated with diabetes and obesity. Ask about previous abdominal surgery.

Drug history

Ask about all prescribed medications, over-the-counter medicines and herbal preparations. Many drugs affect the gastrointestinal tract (Box 6.9) and are hepatotoxic.

Family history

Inflammatory bowel disease is more common in patients with a family history of either Crohn's disease or ulcerative colitis. Colorectal cancer in a first-degree relative increases the risk of colorectal cancer and polyps. Peptic ulcer disease may be familial but is mainly due to environmental factors, such as the transmission of *H. pylori* infection. Gilbert's syndrome is an autosomal dominant condition; haemochromatosis and Wilson's disease are autosomal recessive disorders. Autoimmune diseases, particularly thyroid disease, are common in relatives of those with primary biliary cirrhosis and autoimmune hepatitis. A family history of diabetes is frequently seen in the context of NAFLD.

Social history

Ask about:

- Dietary history: assess the intake of calories and sources of essential nutrients. For guidance, there are 9 kcal per g of fat and 4 kcal per g of carbohydrates and protein.
- Food intolerances: patients with irritable bowel syndrome often report specific food intolerances, including wheat, dairy products and others. Painless diarrhoea may indicate high alcohol intake, lactose intolerance or coeliac disease.
- Alcohol consumption: calculate the patient's intake in units (p. 75).
- Smoking: this increases the risk of oesophageal cancer, colorectal cancer, Crohn's disease and peptic ulcer, while patients with ulcerative colitis are less likely to smoke.
- Stress: many disorders, particularly irritable bowel syndrome and dyspepsia, are exacerbated by stress and mental disorders.
- Foreign travel: this is particularly relevant in liver disease and diarrhoea.
- Risk factors for liver disease: these include intravenous drug use, tattoos, foreign travel, blood transfusions, and sex between men or with prostitutes and multiple sexual partners. Hepatitis B and C may present with chronic liver disease or cancer decades after the primary infection, so enquire about risk factors in the distant as well as the recent past.

The physical examination

Examination of the gastrointestinal system includes intimate examination and consideration should be given to a patient's preference regarding the gender of the examining clinician and the presence of a chaperone. Skin colour should be taken into account when eliciting clinical signs such as jaundice. Differential diagnosis may need to be adjusted to consider racial or ethnic differences.

General examination

Examination sequence (Video 6)



- Note the patient's demeanour and general appearance. Are they in pain, cachectic, thin, well-nourished or obese? Record height, weight, waist circumference and body mass index (p. 32). Note whether obesity is truncal or generalised. Look for abdominal striae or loose skin folds.

- Inspect the patient's hands for clubbing, koilonychia (spoon-shaped nails) and signs of chronic liver disease (Fig. 6.9), including leuconychia (white nails) and palmar erythema.
- Inspect the mouth, throat and tongue.
- Ask the patient to look down and retract the upper eyelid to expose the sclera; look to see if it is yellow in natural light (see Fig. 6.8).
- Examine the cervical, axillary and inguinal lymph nodes (p. 34).

6

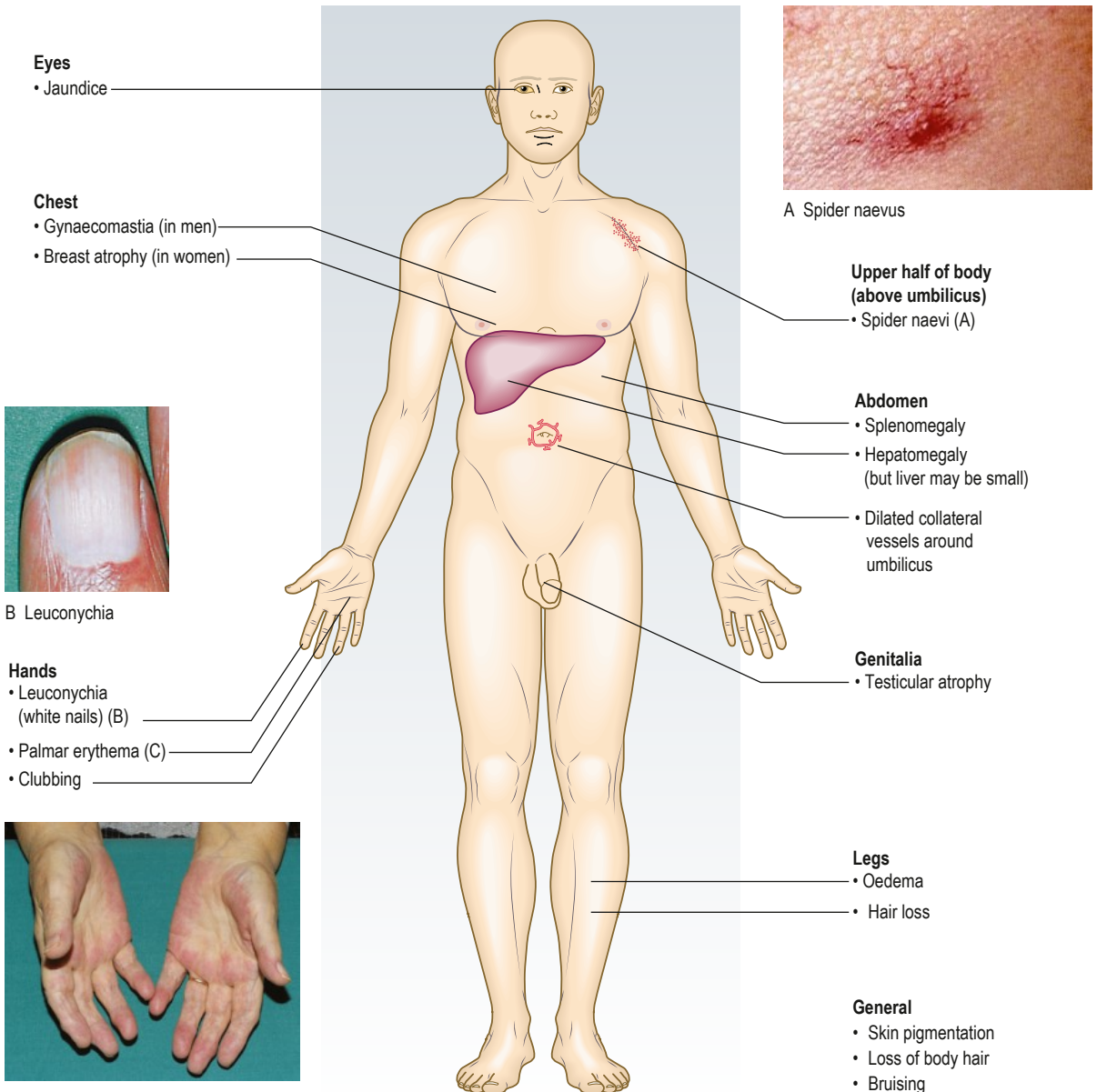


Fig. 6.9 Features of chronic liver disease.

Striae indicate rapid weight gain, previous pregnancy or, rarely, Cushing's syndrome. Loose skin folds signify recent weight loss.

Stigmata of iron deficiency include angular cheilitis (painful cracks at the corners of the mouth) and atrophic glossitis (pale, smooth tongue). The tongue has a beefy, raw appearance in folate and vitamin B₁₂ deficiency. Mouth and throat aphthous ulcers are common in coeliac and inflammatory bowel disease (see Fig. 6.3B).

Gastric and pancreatic cancer may spread to cause enlargement of the left supraclavicular lymph nodes (Troisier's sign). More widespread lymphadenopathy with hepatosplenomegaly suggests lymphoma.

Liver disease

Do not confuse the diffuse yellow sclerae of jaundice with small, yellowish fat pads (pingueculae) sometimes seen at the periphery of the sclerae.

Certain signs (stigmata) suggest chronic liver disease (see Fig. 6.9):

- Palmar erythema and spider naevi are caused by excess oestrogen associated with the reduced hepatic breakdown of sex steroids. Spider naevi are isolated telangiectasias that characteristically fill from a central vessel and are found in the distribution of the superior vena cava (upper trunk, arms and face). Women may have up to five spider naevi in health; palmar erythema and numerous spider naevi are normal during pregnancy. In men, these signs suggest chronic liver disease.
- Gynaecomastia (breast enlargement in males), with loss of body hair and testicular atrophy, may occur due to reduced breakdown of oestrogens.
- Leuconychia, caused by hypoalbuminaemia, may also occur in protein-calorie malnutrition (kwashiorkor), malabsorption due to protein-losing enteropathy, as in coeliac disease, or heavy and prolonged proteinuria (nephrotic syndrome).
- Finger clubbing is found in liver cirrhosis, inflammatory bowel disease and malabsorption syndromes.

Other signs that may be associated with liver disease include:

- Dupuytren's contracture of the palmar fascia (see Fig. 3.5): linked with alcohol-related chronic liver disease
- bilateral parotid swelling due to sialoadenosis: may be a feature of chronic alcohol misuse.

Signs that suggest liver failure include:

- asterixis, a coarse flapping tremor when the arms are outstretched and hands dorsiflexed, which occurs with hepatic encephalopathy
- fetor hepaticus, a distinctive 'mousy' odour of dimethyl sulphide on the breath, which is evidence of portosystemic shunting (with or without encephalopathy)
- altered mental state, varying from drowsiness with the day/night pattern reversed, through confusion and disorientation, to unresponsive coma

- jaundice, yellowing of the skin may be less noticeable in patients with black or brown skin
- ascites
- late neurological features, which include spasticity, an extension of the arms and legs, and extensor plantar responses.

In a jaundiced patient, spider naevi, palmar erythema and ascites all strongly suggest chronic liver disease rather than obstructive jaundice.

Abdominal examination

Examine the patient in good light and warm surroundings, positioned comfortably supine, arms by the side with the head resting on only one or two pillows to relax the abdominal wall muscles. Use extra pillows to support a patient with kyphosis or breathlessness.

Inspection

Examination sequence (Video 6A)



- Look at the teeth, tongue and buccal mucosa; check for mouth ulcers.
- Note any smells, including alcohol, fetor hepaticus, uraemia, melaena or ketones.
- Expose the abdomen from the xiphisternum to the symphysis pubis, leaving the chest and legs covered.

The normal abdomen is flat or slightly scaphoid and symmetrical. At rest, respiration is principally diaphragmatic; the abdominal wall moves out and the liver, spleen and kidneys move downwards during inspiration. The umbilicus is usually inverted.

Skin

In older patients, seborrhoeic warts, ranging from pink to brown or black, and haemangiomas (Campbell de Morgan spots) are common and normal, but note any striae, bruising or scratch marks.

Visible veins

Abnormally prominent veins on the abdominal wall suggest portal hypertension or vena cava obstruction. In portal hypertension, recanalisation of the umbilical vein along the falciform ligament produces distended veins that drain away from the umbilicus: the 'caput medusae'. The umbilicus may appear bluish and distended due to an umbilical varix. In contrast, an umbilical hernia is a distended and everted umbilicus that does not appear vascular and may have a palpable cough impulse. Dilated tortuous veins with blood flow superiorly are collateral veins caused by obstruction of the inferior vena cava. Rarely, superior vena cava obstruction gives rise to similarly distended abdominal veins, but these all flow inferiorly.

Abdominal swelling

Diffuse abdominal swelling could be due to ascites or intestinal obstruction. If localised, it could be caused by urinary retention, a mass or an enlarged organ such as the liver. In obesity, the umbilicus is usually sunken; in ascites, it is flat or, more commonly, everted. Look tangentially across the abdomen and from the foot of the bed for any asymmetry suggesting a localised mass.

Abdominal scars and stomas

Note any surgical scars or stomas and clarify what operations have been undertaken (Figs 6.10 and 6.11). A small infraumbilical incision usually indicates a previous laparoscopy. Puncture scars from laparoscopic surgical ports may be visible. An incisional hernia at the site of a scar is palpable as a defect in the abdominal wall musculature and becomes more obvious as the patient raises their head off the bed or coughs.

Palpation

Examination sequence (Video 6B)

- Ensure your hands are warm and clean.
- If the bed is low, kneel beside it but avoid touching the floor to prevent infection.
- Ask the patient to show you where any pain is and to report any tenderness during palpation.
- Ask the patient to place their arms by their sides to help relax the abdominal wall.
- Use your right hand, keeping it flat and in contact with the abdominal wall.

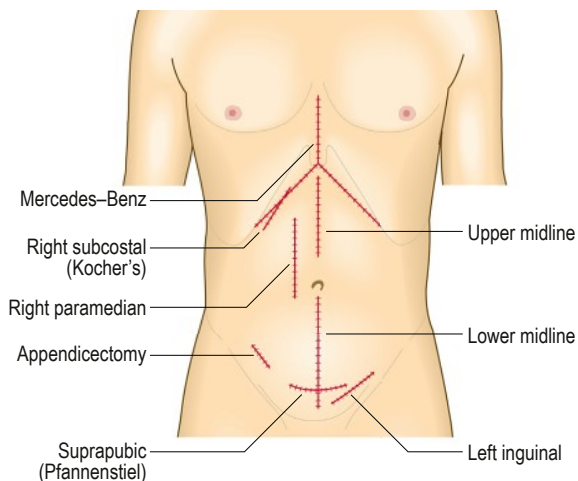


Fig. 6.10 Some abdominal incisions. The midline and oblique incisions avoid damage to innervation of the abdominal musculature and later development of incisional hernias. These incisions have been widely superseded by laparoscopic surgery, however.

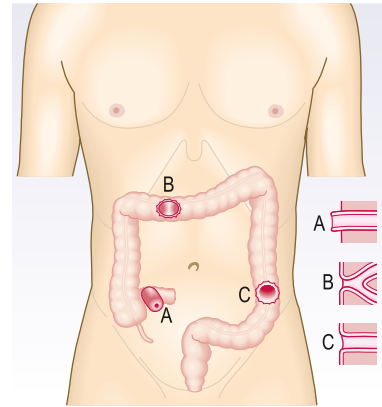


Fig. 6.11 Surgical stomas. **A** An ileostomy is usually in the right iliac fossa and is formed as a spout. **B** A loop colostomy is created to defunction the distal bowel temporarily. It is usually in the transverse colon and has afferent and efferent limbs. **C** A colostomy may be terminal: that is, resected distal bowel. It is usually flush and in the left iliac fossa.

- Observe the patient's face throughout for any sign of discomfort.
- Begin with light superficial palpation away from any site of pain.
- Palpate each region in turn, and then repeat with deeper palpation.
- Test abdominal muscle tone using light, dipping finger movements.
- Describe any mass using the basic principles outlined in Chapter 3, p. 34. Describe its site, size, surface, shape and consistency, and note whether it moves on respiration. Is the mass fixed or mobile?
- To determine if a mass is superficial and in the abdominal wall rather than within the abdominal cavity, ask the patient to tense their abdominal muscles by lifting their head. An abdominal wall mass will still be palpable, whereas an intra-abdominal mass will not.
- Decide whether the mass is an enlarged abdominal organ or separate from the solid organs.

Tenderness

Discomfort during palpation may vary and may be accompanied by resistance to palpation. Consider the patient's level of anxiety when assessing the severity of pain and degree of tenderness elicited. Tenderness in several areas on minimal pressure may be due to generalised peritonitis but is more often caused by anxiety. Severe superficial pain with no tenderness on deep palpation or pain that disappears if the patient is distracted also suggests anxiety. With these exceptions, tenderness usefully indicates underlying pathology.

Voluntary guarding is the voluntary contraction of the abdominal muscles when palpation provokes pain. Involuntary guarding is the reflex contraction of the abdominal muscles when there is inflammation of the parietal peritoneum. If the whole peritoneum is inflamed (generalised peritonitis) due to a

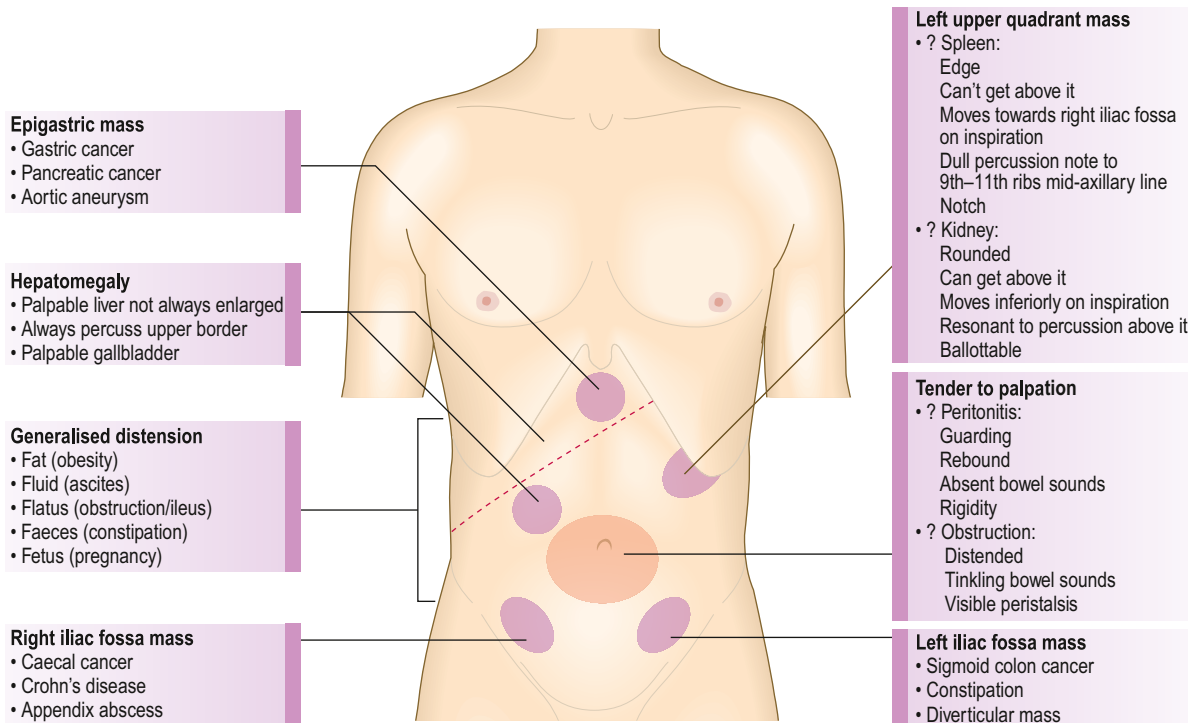


Fig. 6.12 Palpable abnormalities in the abdomen.

6.10 Specific signs in the 'acute abdomen'

Sign	Disease associations	Examination
Murphy's	Acute cholecystitis: Sensitivity 50–97% Specificity 50–80%	As the patient takes a deep breath in, gently palpate in the right upper quadrant of the abdomen; the acutely inflamed gallbladder contacts the examining fingers, evoking pain with the arrest of inspiration
Rovsing's	Acute appendicitis: Sensitivity 20–70% Specificity 40–96%	Palpation in the left iliac fossa produces pain in the right iliac fossa
Iliopsoas	Retroileal appendicitis, iliopsoas abscess, perinephric abscess	Ask the patient to flex their thigh against the resistance of your hand; a painful response indicates an inflammatory process involving the right psoas muscle
Grey Turner's and Cullen's	Haemorrhagic pancreatitis, aortic rupture and ruptured ectopic pregnancy (see Fig. 6.13)	Bleeding into the falciform ligament; bruising develops around the umbilicus (Cullen) or in the loins (Grey Turner)

perforated viscus, the abdominal wall no longer moves with respiration; breathing becomes increasingly thoracic and the anterior abdominal wall muscles are held rigid (board-like rigidity).

The site of tenderness is important. Tenderness in the epigastrium suggests peptic ulcer; in the right hypochondrium, cholecystitis; in the left iliac fossa, diverticulitis; and in the right iliac fossa, appendicitis or Crohn's ileitis (Fig. 6.12). Ask the patient to cough or gently percuss the abdomen to elicit any pain or 'rebound tenderness'. This approach is preferable to rapidly removing your hand after deep palpation. Rebound tenderness is

a sign of intra-abdominal disease but not necessarily of parietal peritoneal inflammation (peritonism). Specific abdominal signs are shown in Box 6.10; signs of Grey Turner's and Cullen's are shown in Fig. 6.13. Typical findings may be masked in patients taking glucocorticoids, immunosuppressants or anti-inflammatory drugs, in alcohol intoxication or altered states of consciousness.

Palpable mass

A pulsatile mass palpable in the upper abdomen may be normal aortic pulsation in a thin person, a gastric or pancreatic tumour transmitting underlying aortic pulsation or an aortic aneurysm.

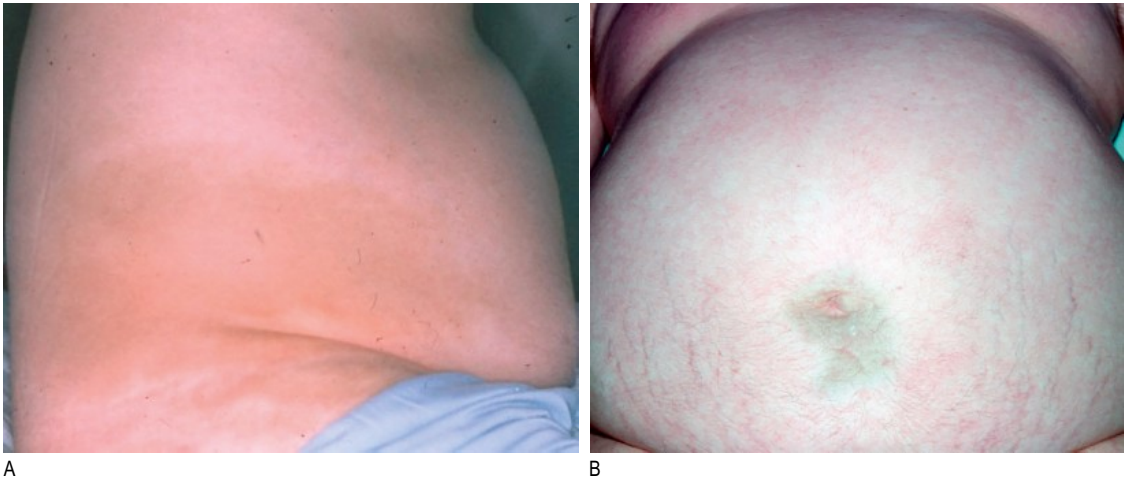


Fig. 6.13 Acute pancreatitis. [A] Bruising over the flanks (Grey Turner's sign). [B] Bruising around the umbilicus (Cullen's sign).

A pathological mass can usually be distinguished from normal palpable structures by site (Fig. 6.14), and from palpable faeces as these can be indented and may disappear following defecation. A hard subcutaneous nodule at the umbilicus may indicate metastatic cancer ('Sister Mary Joseph's nodule').

Enlarged organs

Examine the liver, gallbladder, spleen and kidneys in turn during deep inspiration. Keep your examining hand still and wait for the organ to move with breathing. Do not start palpation too close to the costal margin, missing the edge of the liver or spleen.

Hepatomegaly

Examination sequence (Video 6C)



- Place your hand flat on the skin of the right iliac fossa.
- Point your fingers upwards and your index and middle fingers lateral to the rectus muscle, so that your fingertips lie parallel to the rectus sheath (Fig. 6.15). Keep your hand stationary.
- Ask the patient to breathe in deeply through the mouth.
- Feel for the liver edge as it descends on inspiration.
- Move your hand progressively up the abdomen, 1 cm at a time, between each breath the patient takes until you reach the costal margin or detect the liver edge.
- If you feel a liver edge, describe:
 - size
 - surface: smooth or irregular
 - edge: smooth or irregular; define the medial border
 - consistency: soft or hard
 - tenderness
 - pulsatility.
- To examine for gallbladder tenderness, ask the patient to breathe in deeply, then gently palpate the right upper quadrant in the mid-clavicular line.

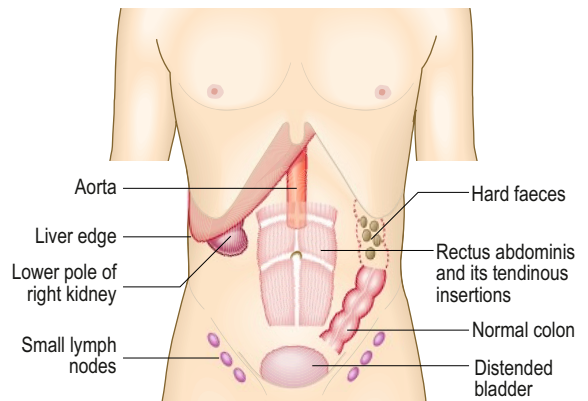


Fig. 6.14 Palpable masses that may be physiological rather than pathological.

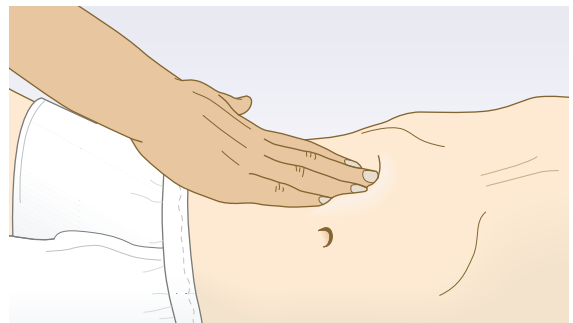


Fig. 6.15 Palpation of the liver.

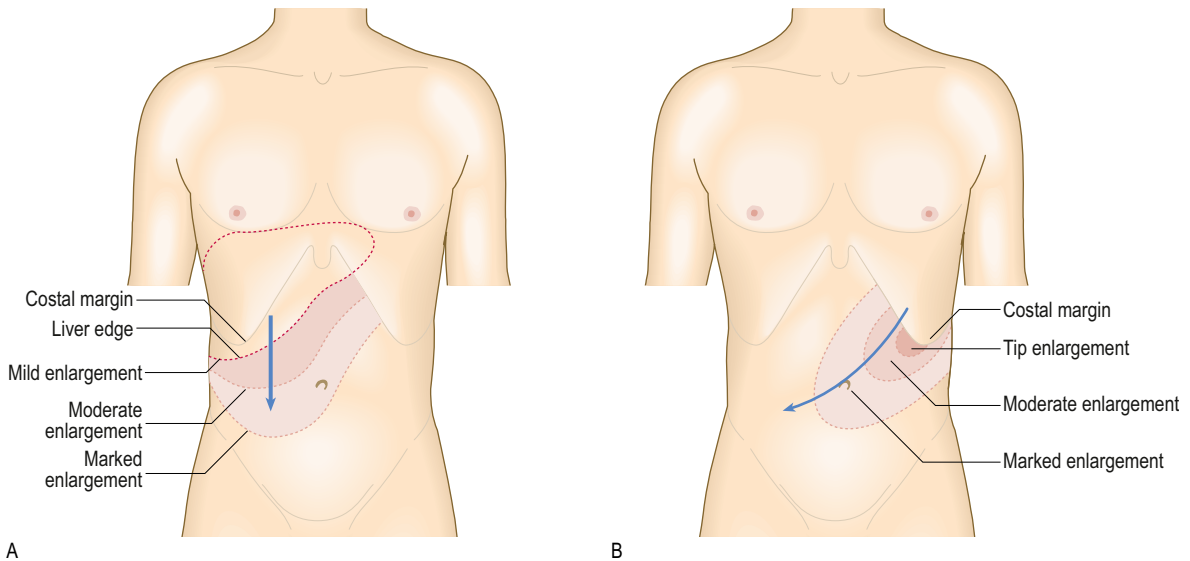


Fig. 6.16 Patterns of progressive enlargement of the liver and spleen. **A** Direction of enlargement of the liver. **B** Direction of enlargement of the spleen. The spleen moves downwards and medially during inspiration.

Percussion

Examination sequence

- Ask the patient to hold their breath in full expiration.
- Percuss downwards from the right fifth intercostal space in the mid-clavicular line, listening for dullness indicating the upper border of the liver.
- Measure the distance in centimetres below the costal margin in the mid-clavicular line or from the upper border of dullness to the palpable liver edge.

In the normal abdomen, you may feel:

- the liver edge below the right costal margin
- the aorta as a pulsatile swelling above the umbilicus
- the lower pole of the right kidney in the right flank
- faecal scybala (hardened masses of faeces) in the sigmoid colon in the left iliac fossa
- a full bladder arising out of the pelvis in the suprapubic region.

The normal liver is identified as an area of dullness to percussion over the right anterior chest between the fifth rib and the costal margin.

The liver may be enlarged (Fig. 6.16A) or displaced downwards by hyperinflated lungs.

Hepatic enlargement can result from chronic parenchymal liver disease from any cause (Box 6.11). The liver is enlarged in early cirrhosis but often shrunken in advanced cirrhosis. Fatty liver (hepatic steatosis) can cause marked hepatomegaly. Hepatic enlargement due to metastatic tumour is hard and irregular. An enlarged left lobe may be felt in the epigastrium or even

the left hypochondrium. In right heart failure, the congested liver is usually soft and tender; a pulsatile liver indicates tricuspid regurgitation. A bruit over the liver may be heard in acute alcoholic hepatitis, hepatocellular cancer, arteriovenous malformation and in patients with portal-systemic shunts. The most common reason for an audible bruit over the liver, however, is a transmitted heart murmur. Liver failure produces additional symptoms of encephalopathy, which can be graded (Box 6.12).

Resonance below the fifth intercostal space suggests hyperinflated lungs or occasionally the interposition of the transverse colon between the liver and the diaphragm (Chilaiditi's sign).

6.11 Causes of hepatomegaly

Chronic parenchymal liver disease

- Alcoholic liver disease
- Hepatic steatosis
- Autoimmune hepatitis
- Viral hepatitis
- Primary biliary cirrhosis

Malignancy

- Primary hepatocellular cancer
- Secondary metastatic cancer

Right heart failure

Haematological disorders

- Lymphoma
- Leukaemia
- Myelofibrosis
- Polycythaemia

Rarities

- Amyloidosis
- Budd–Chiari syndrome
- Sarcoidosis
- Glycogen storage disorders

6.12 Grading of hepatic encephalopathy (West Haven)

Stage	State of consciousness
0	No change in personality or behaviour No asterixis (flapping tremor)
1	Impaired concentration and attention span Sleep disturbance, slurred speech Euphoria or depression Asterixis present
2	Lethargy, drowsiness, apathy or aggression Disorientation, inappropriate behaviour, slurred speech
3	Confusion and disorientation, bizarre behaviour Drowsiness or stupor Asterixis usually absent
4	Comatose with no response to voice commands Minimal or absent response to painful stimuli
<i>Reproduced from Conn HO, Leevy CM, Vlahcevic ZR, et al. Comparison of lactulose and neomycin in the treatment of chronic portal-systemic encephalopathy. A double-blind controlled trial. Gastroenterology. 1977;72(4):573, with permission from Elsevier Inc.</i>	

In a patient with right upper quadrant pain, test for Murphy's sign (see [Box 6.10](#)); a positive test modestly increases the probability of acute cholecystitis. Palpable distension of the gallbladder is rare and has a characteristic globular shape. It results from either obstruction of the cystic duct, as in mucocoele or empyema of the gallbladder, or obstruction of the common bile duct with a patent cystic duct, as in pancreatic cancer. In a jaundiced patient, a palpable gallbladder is likely to be due to extrahepatic obstruction, such as from pancreatic cancer or, very rarely, gallstones (Courvoisier's sign). In gallstone disease, the gallbladder may be tender but impalpable because of fibrosis of the gallbladder wall.

Splenomegaly

The spleen has to enlarge threefold before it becomes palpable, so a palpable spleen always indicates splenomegaly. It enlarges from under the left costal margin down and medially towards the umbilicus ([Fig. 6.16B](#)). A characteristic notch may be palpable midway along its leading edge, helping differentiate it from an enlarged left kidney ([Box 6.13](#)).

Examination sequence (Video 6D)

- Place your hand over the patient's umbilicus. With your hand stationary, ask the patient to inhale deeply through the mouth.
- Feel for the splenic edge as it descends on inspiration.
- Move your hand diagonally upwards towards the left hypochondrium ([Fig. 6.17A](#)), 1 cm at a time between each breath the patient takes.
- Feel the costal margin along its length, as the position of the spleen tip is variable.

6.13 Differentiating a palpable spleen from the left kidney

Distinguishing feature	Spleen	Kidney
Mass is smooth and regular in shape	More likely	Polycystic kidneys are bilateral irregular masses
Mass descends in inspiration	Yes, travels superficially and diagonally	Yes, moves deeply and vertically
Ability to feel deep into the mass	Yes	No
Palpable notch on the medial surface	Yes	No
Bilateral masses palpable	No	Sometimes (e.g. polycystic kidneys)
Percussion resonant over the mass	No	Sometimes
Mass extends beyond the midline	Sometimes	No (except with horseshoe kidney)

- If you cannot feel the splenic edge, palpate with your right hand, placing your left hand behind the patient's left lower ribs and pulling the ribcage forward ([Fig. 6.17B](#)), or ask the patient to roll towards you and on to their right side and repeat the above.
- Feel along the left costal margin and percuss over the lateral chest wall. The normal spleen causes dullness to percussion posterior to the left mid-axillary line beneath the 9th–11th ribs.

There are many causes of splenomegaly ([Box 6.14](#)). Massive enlargement in the developed world is usually due to myeloproliferative disease or haematological malignancy; worldwide, tropical splenomegaly syndrome due to malaria is a common cause.

Important causes of hepatosplenomegaly include lymphoma or myeloproliferative disorders, cirrhosis with portal hypertension, amyloidosis, sarcoidosis and glycogen storage disease.

Ascites

Ascites is the accumulation of intraperitoneal fluid (see [Fig. 6.6](#)).

Examination sequence (Video 6E)**Shifting dullness**

- With the patient supine, percuss from the midline out to the flanks ([Fig. 6.18](#)). Note any change from resonant to dull, along with areas of dullness and resonance.
- Keep your finger on the site of dullness in the flank and ask the patient to turn on to their opposite side.
- Pause for 10 seconds to allow any ascites to gravitate, then percuss again. If the area of dullness is now resonant, shifting dullness is present, indicating ascites.

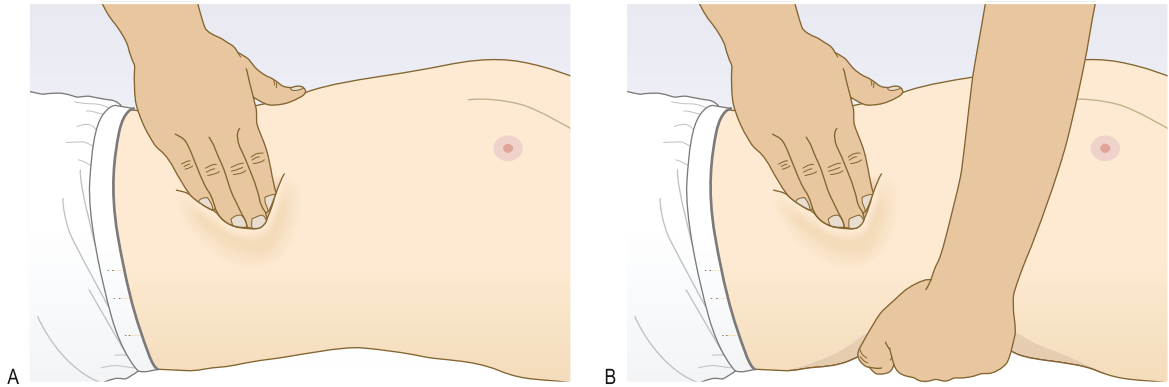


Fig. 6.17 Palpation of the spleen. **A** Initial palpation for the splenic edge moving diagonally from the umbilicus to the left hypochondrium. **B** If the spleen is impalpable by the method shown in A, use your left hand to pull the ribcage forward and elevate the spleen, making it more likely to be palpable by your right hand.

6.14 Causes of splenomegaly

Haematological disorders

- Lymphoma and lymphatic leukaemias
- Myeloproliferative diseases, polycythaemia rubra vera and myelofibrosis
- Haemolytic anaemia, congenital spherocytosis

Portal hypertension

Infections

- Glandular fever
- Malaria, kala-azar (leishmaniasis)
- Bacterial endocarditis
- Brucellosis, tuberculosis, salmonellosis

Rheumatological conditions

- Rheumatoid arthritis (Felty's syndrome)
- Systemic lupus erythematosus

Rarities

- Sarcoidosis
- Amyloidosis
- Glycogen storage disorders

Fluid thrill

- If the abdomen is tensely distended and you are uncertain whether ascites is present, feel for a fluid thrill.
- Place the palm of your left hand flat against the left side of the patient's abdomen and flick a finger of your right hand against the right side of the abdomen.
- If you feel a ripple against your left hand, ask an assistant or the patient to place the edge of their hand on the midline of the abdomen (Fig. 6.19). This prevents transmission of the impulse via the skin rather than through the ascites. If you still feel a ripple against your left hand, a fluid thrill is present (detected only in gross ascites).

The causes of ascites are shown in Box 6.15.

Auscultation

Examination sequence (Video 6F)

- With the patient supine, place your stethoscope diaphragm to the right of the umbilicus and do not move it.

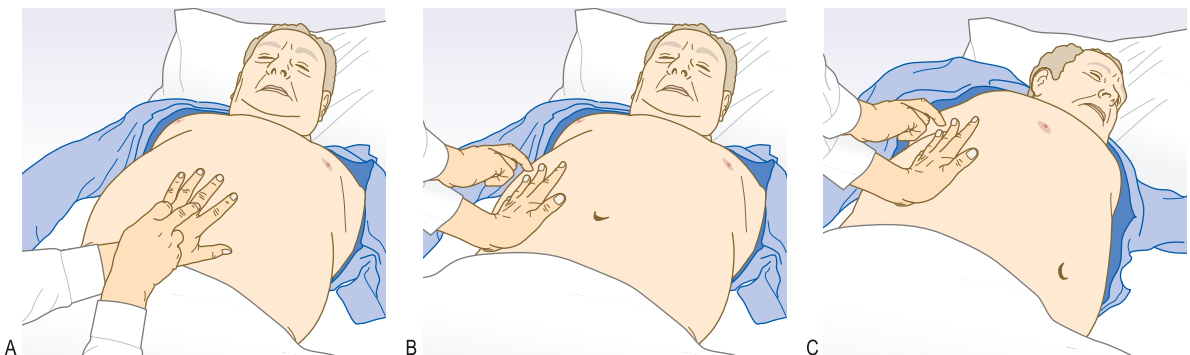


Fig. 6.18 Percussing for ascites. **A** and **B** Percuss towards the flank from resonant to dull. **C** Then ask the patient to roll onto their other side. In ascites the note then becomes resonant.

- Listen for up to 2 minutes before concluding that bowel sounds are absent.
- Listen above the umbilicus over the aorta for arterial bruits.
- Now listen 2–3 cm above and lateral to the umbilicus for bruits from renal artery stenosis.
- Listen over the liver for bruits.
- Test for a succussion splash; this sounds like a half-filled water bottle being shaken. Explain the procedure to the patient, then shake their abdomen by rocking their pelvis using both hands.

Normal bowel sounds are gurgling noises from the normal peristaltic activity of the gut. They normally occur every 5–10 seconds but the frequency varies.

The absence of bowel sounds implies paralytic ileus or peritonitis. In intestinal obstruction, bowel sounds occur with increased frequency and volume and have a high-pitched,

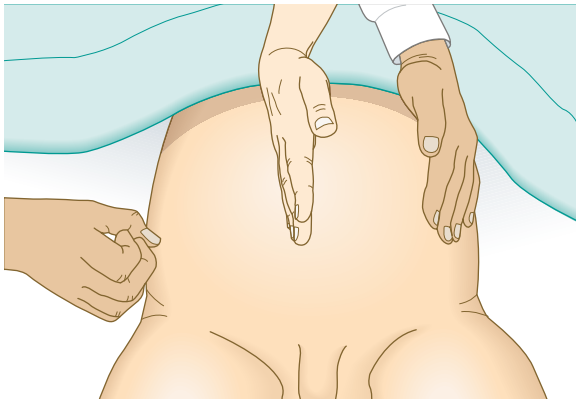


Fig. 6.19 Eliciting a fluid thrill.

6.15 Causes of ascites

Diagnosis	Comment
Common	
Hepatic cirrhosis with portal hypertension	Transudate
Intra-abdominal malignancy with peritoneal spread	Exudate, cytology may be positive
Uncommon	
Hepatic vein occlusion (Budd–Chiari syndrome)	Transudate in the acute phase
Constrictive pericarditis and right heart failure	Check jugular venous pressure and listen for pericardial rub
Hypoproteinaemia (nephrotic syndrome, protein-losing enteropathy)	Transudate
Tuberculous peritonitis	Low glucose content
Pancreatitis, pancreatic duct disruption	Very high amylase content

tinkling quality. Bruits suggest an atheromatous or aneurysmal aorta or superior mesenteric artery stenosis. A friction rub, which sounds like rubbing your dry fingers together, may be heard over the liver (perihepatitis) or spleen (perisplenitis). An audible splash more than 4 hours after the patient has eaten or drunk anything indicates delayed gastric emptying, as in pyloric stenosis.

Hernias

The inguinal canal extends from the pubic tubercle to the anterior superior iliac spine (Fig. 6.20). It has an internal ring at the mid-inguinal point (midway between the pubic symphysis and the anterior superior iliac spine) and an external ring at the pubic tubercle. The femoral canal lies below the inguinal ligament and lateral to the pubic tubercle.

Hernias are common and typically occur at openings of the abdominal wall, such as the inguinal, femoral and obturator canals, the umbilicus and the oesophageal hiatus. They may also occur at sites of the weakness of the abdominal wall, as in previous surgical incisions.

An external abdominal hernia is an abnormal protrusion of the bowel and/or omentum from the abdominal cavity. External hernias are more obvious when the pressure within the abdomen rises, such as when the patient is standing, coughing or straining at stool. Internal hernias occur through defects of the mesentery or into the retroperitoneal space and are not visible.

An impulse can often be felt in a hernia during coughing (cough impulse). Identify a hernia from its anatomical site and characteristics and attempt to differentiate between direct and indirect inguinal hernias.

Examination sequence

- Examine the groin with the patient standing upright.

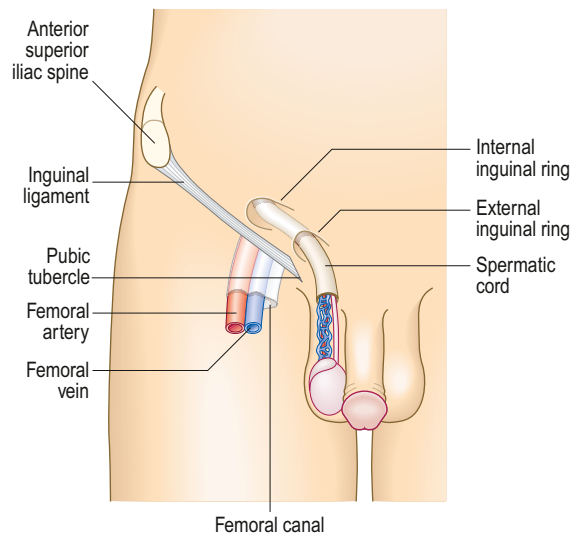


Fig. 6.20 Anatomy of the inguinal canal and femoral sheath.

- Inspect the inguinal and femoral canals and the scrotum for any lumps or bulges.
- Ask the patient to cough; look for an impulse over the femoral or inguinal canal and scrotum.
- Identify the anatomical relationships between the bulge, the pubic tubercle and the inguinal ligament to distinguish a femoral from an inguinal hernia.
- Palpate the external inguinal ring and along the inguinal canal for possible muscle defects. Ask the patient to cough and feel for a cough impulse.
- Now ask the patient to lie down and establish whether the hernia reduces spontaneously.
- If so, press two fingers over the internal inguinal ring at the mid-inguinal point and ask the patient to cough or stand up while you maintain pressure over the internal inguinal ring. If the hernia reappears, it is a direct hernia. If it can be prevented from reappearing, it is an indirect inguinal hernia.
- Examine the opposite side to exclude the possibility of asymptomatic hernias.

An indirect inguinal hernia bulges through the internal ring and follows the course of the inguinal canal. It may extend beyond the external ring and enter the scrotum. Indirect hernias comprise 85% of all hernias and are more common in younger men.

A direct inguinal hernia forms at a site of muscle weakness in the posterior wall of the inguinal canal and rarely extends into the scrotum. It is more common in older men and women (Fig. 6.21).

A femoral hernia projects through the femoral ring and into the femoral canal. Inguinal hernias are palpable above and medial to the pubic tubercle. Femoral hernias are palpable below the inguinal ligament and lateral to the pubic tubercle.

In a reducible hernia, the contents can be returned to the abdominal cavity, spontaneously or by manipulation; if they cannot, the hernia is irreducible. An abdominal hernia has a covering sac of the peritoneum and the neck of the hernia is a common site of compression of the contents (Fig. 6.22). If the hernia contains bowel, obstruction may occur. If the blood supply to the contents of the hernia (bowel or omentum) is restricted, the hernia is strangulated. It is tense, tender and has no cough impulse, there may be bowel obstruction and, later,



Fig. 6.21 Right inguinal hernia.

signs of sepsis and shock. A strangulated hernia is a surgical emergency and, if left untreated, will lead to bowel infarction and peritonitis.

Rectal examination

Digital examination of the rectum is important (Box 6.16). Do not avoid it because you or the patient find it disagreeable. The patient's verbal consent is needed, however, and the examination should be carried out in the presence of a chaperone.

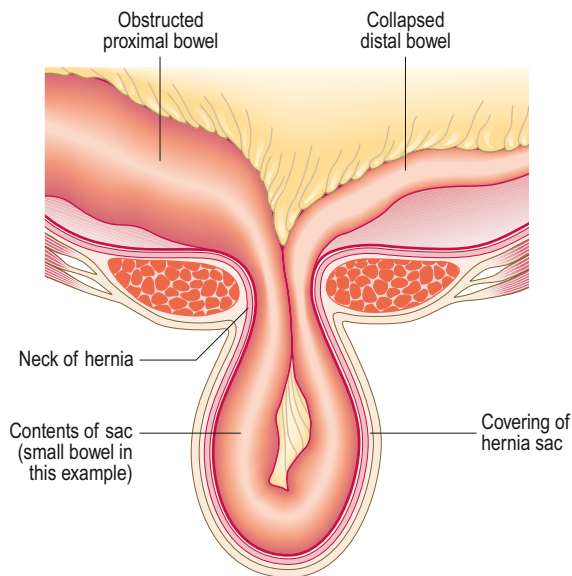


Fig. 6.22 Hernia: anatomical structure.

6.16 Indications for rectal examination

Alimentary

- Suspected appendicitis, pelvic abscess, peritonitis, lower abdominal pain
- Diarrhoea, constipation, tenesmus or anorectal pain
- Rectal bleeding or iron deficiency anaemia
- Unexplained weight loss
- Bimanual examination of lower abdominal mass for diagnosis or staging
- Malignancies of unknown origin

Genitourinary

- Assessment of prostate in prostatism or suspected prostatic cancer
- Dysuria, frequency, haematuria, epididymo-orchitis
- Replacement for vaginal examination when this would be inappropriate

Miscellaneous

- Unexplained bone pain, backache or lumbosacral nerve root pain
- Pyrexia of unknown origin
- Abdominal, pelvic or spinal trauma

The normal rectum is usually empty and smooth-walled, with the coccyx and sacrum lying posteriorly. In the male, anterior to the rectum from below upwards, lie the membranous urethra, the prostate and the base of the bladder. The normal prostate is smooth and firm, with lateral lobes and a median groove between them. In the female, the vagina and cervix lie anteriorly. The upper end of the anal canal is marked by the puborectalis muscle, which is readily palpable and contracts as a reflex action on coughing or conscious contraction by the patient. Beyond the anal canal, the rectum passes upwards and backwards along the curve of the sacrum.

Spasm of the external anal sphincter is common in anxious patients. When associated with local pain, it is probably due to an anal fissure (a mucosal tear). If you suspect an anal fissure, give the patient a local anaesthetic suppository 10 minutes before the examination to reduce the pain and spasm, and to aid examination.

Examination sequence

- Explain what you are going to do and why it is necessary and ask for permission to proceed. Tell the patient that the examination may be uncomfortable but should not be painful.
- Offer a chaperone; record a refusal. Make a note of the name of the chaperone.
- Position the patient in the left lateral position with their buttocks at the edge of the couch, their knees drawn up to their chest and their heels clear of the perineum (Fig. 6.23).
- Put on gloves and examine the perianal skin, using an effective light source.
- Look for skin lesions, external haemorrhoids, fissures and fistulae.
- Lubricate your index finger with water-based gel.
- Place the pulp of your forefinger on the anal margin and apply steady pressure on the sphincter to push your finger gently through the anal canal into the rectum (Fig. 6.24).
- If an anal spasm occurs, ask the patient to breathe in deeply and relax. If necessary, use a local anaesthetic suppository or

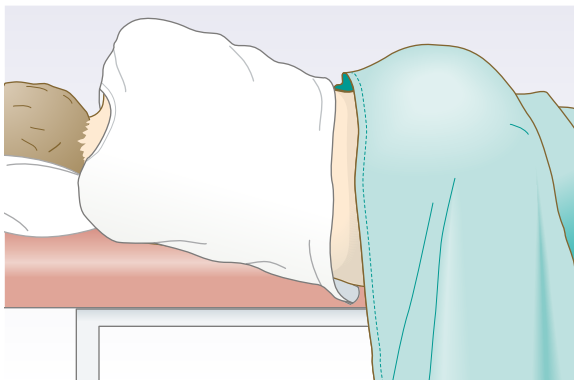


Fig. 6.23 The correct position of the patient before a rectal examination.

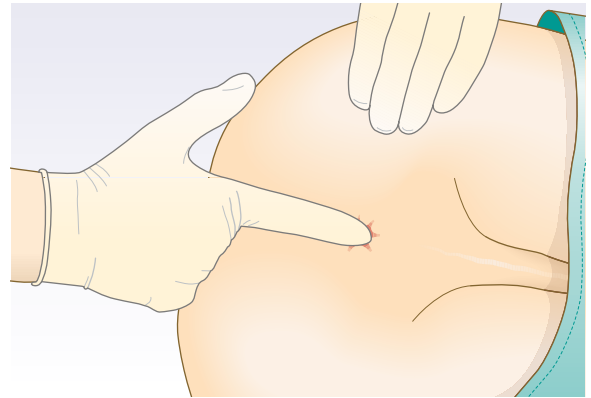


Fig. 6.24 Rectal examination. The correct method for inserting your index finger in rectal examination.

- gel before trying again. If pain persists, examination under general anaesthesia may be necessary.
- Ask the patient to squeeze your finger with their anal muscles and note any weakness of sphincter contraction.
- Palpate systematically around the entire rectum; note any abnormality and examine any mass (Fig. 6.25). Record the percentage of the rectal circumference involved by disease and its distance from the anus.
- Identify the uterine cervix in women and the prostate in men; assess the size, shape and consistency of the prostate and note any tenderness.
- If the rectum contains faeces and you are in doubt about palpable masses, repeat the examination after the patient has defecated.
- Slowly withdraw your finger. Examine it for stool colour and the presence of blood or mucus (Box 6.17).

Haemorrhoids ('piles', congested venous plexuses around the anal canal) are usually palpable if thrombosed. In patients with chronic constipation, the rectum is often loaded with faeces. Faecal masses are frequently palpable, should be movable and can be indented. In women, a retroverted uterus and the normal cervix are often palpable through the anterior rectal wall and a vaginal tampon may be confusing. Cancer of the lower rectum is palpable as a mucosal irregularity. Obstructing cancer of the upper rectum may produce ballooning of the empty rectal cavity below. Metastases or colonic tumours within the pelvis may be mistaken for faeces and vice versa. Lateralised tenderness suggests pelvic peritonitis. Gynaecological malignancy may cause a 'frozen pelvis' with a hard, rigid feel to the pelvic organs due to extensive peritoneal disease, such as post-radiotherapy or in metastatic cervical or ovarian cancer.

Benign prostatic hyperplasia often produces palpable symmetrical enlargement, but not if the hyperplasia is confined to the median lobe. A hard, irregular or asymmetrical gland with no palpable median groove suggests prostate cancer. Tenderness accompanied by a change in the consistency of the gland may

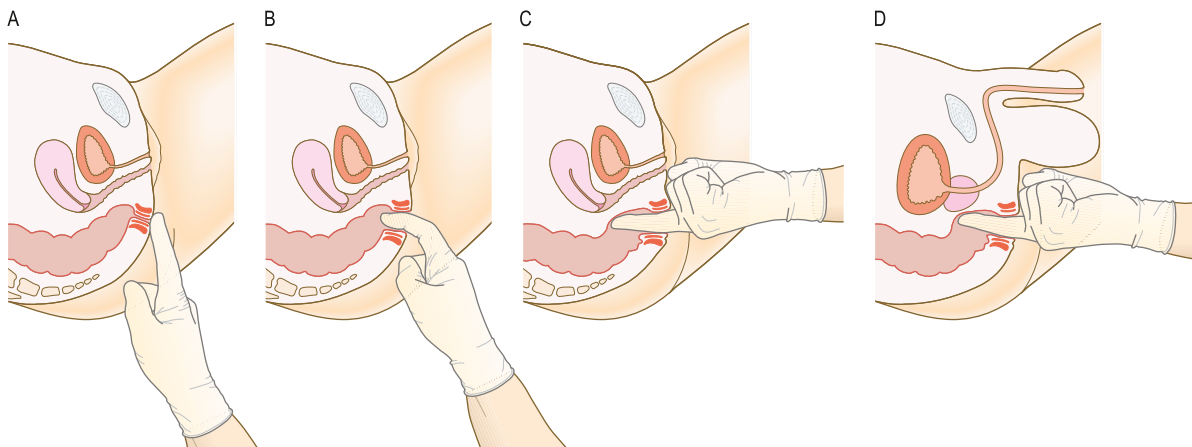


Fig. 6.25 Examination of the rectum. **A** and **B** Insert your finger, then rotate your hand. **C** The most prominent feature in the female is the cervix. **D** The most prominent feature in the male is the prostate.

6.17 Causes of abnormal stool appearance

Stool appearance	Cause
Abnormally pale	Biliary obstruction
Pale and greasy	Steatorrhoea
Black and tarry (melaena)	Bleeding from the upper gastrointestinal tract
Grey/black	Oral iron or bismuth therapy
Silvery	Steatorrhoea plus upper gastrointestinal bleeding (e.g. pancreatic cancer)
Fresh blood in or in stool	Large bowel, rectal or anal bleeding
Stool mixed with pus	Infective colitis or inflammatory bowel disease
Rice-water stool (watery with mucus and cell debris)	Cholera

be caused by prostatitis or prostatic abscess. The prostate is abnormally small in hypogonadism.

Proctoscopy

Proctoscopy is the visual examination of the anal canal; it is an invasive procedure and should only be practised after appropriate training. Always undertake a digital rectal examination first. If examination of the rectal mucosa is required, perform flexible sigmoidoscopy rather than proctoscopy.

Examination sequence

- Place the patient in the left lateral position, as for digital rectal examination.
- With gloved hands, separate the buttocks with the forefinger and thumb of one hand. With your other hand, gently insert a lubricated proctoscope with its obturator in place into the anal canal and rectum in the direction of the umbilicus.
- Remove the obturator and carefully examine the anal canal under good illumination, noting any abnormality. Check for fissures, particularly if the patient reports pain during the procedure.
- Ask the patient to strain down as you slowly withdraw the instrument to detect any degree of rectal prolapse and the presence and severity of any haemorrhoids.

Proctoscopic examination of the anus and lower rectum can confirm or exclude the presence of haemorrhoids, anal fissures and rectal prolapse. Rectal mucosa looks like buccal mucosa, apart from the presence of prominent submucosal veins. During straining, haemorrhoids distend with blood and may prolapse. If the degree of protrusion is more than 3–4 cm, a rectal prolapse may be present.

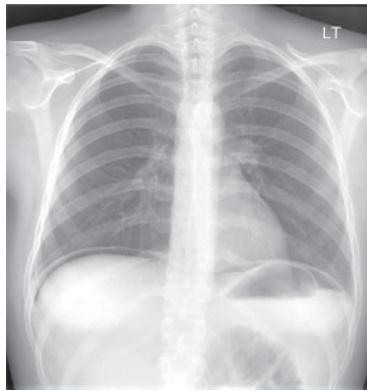
Investigations

Selecting the relevant investigation depends on the clinical problem revealed on history and examination. Investigations are costly and many carry risks, so choose tests capable of distinguishing the likely diagnoses and prioritise the most decisive ones (Box 6.18 and Figs 6.26–6.30).

6.18 Investigations in gastrointestinal and hepatobiliary disease

Investigation	Indication/comment
Clinical samples	
Stool:	
Faecal occult blood by quantitative	Gastrointestinal haemorrhage; sensitive but not specific; used as a population screening tool for colorectal cancer
Faecal Immunochemical Test (qFIT)	Dyspepsia
Faecal H pylori antigen test	Inflammatory bowel disease—raised
Faecal calprotectin	Jaundice (see Box 6.8)
Urine: dipstick or biochemistry	Acute abdominal pain
Ascitic fluid: diagnostic tap	Clear/straw-coloured—normal
	Uniformly blood-stained—malignancy
	Turbid—infection
	Chylous—lymphatic obstruction
	High protein (exudate) —inflammation or malignancy
	Low protein (transudate) —cirrhosis and portal hypertension
Radiology	
Chest X-ray	Suspected acute abdomen, suspected perforated viscus or subphrenic abscess
	Pneumonia, free air beneath diaphragm, pleural effusion, elevated diaphragm
Abdominal X-ray	Intestinal obstruction, perforation, renal colic
	Fluid levels, air above the liver, urinary tract stones
Barium swallow and meal	Only indicated when gastroscopy is not possible and there is suspicion of oesophageal dysmotility, or pharyngeal or gastric outlet obstruction on clinical symptoms (dysphagia or vomiting)
	Oesophageal obstruction (endoscopy preferable, especially if previous gastric surgery)
Small bowel follow-through	Subacute small bowel obstruction, duodenal diverticulosis
Small bowel magnetic resonance imaging or magnetic resonance enteroclysis (real-time imaging of liquid moving through the small bowel)	Crohn's disease, lymphoma, obscure gastrointestinal bleeding
CT colonography	Altered bowel habit, iron deficiency anaemia, rectal bleeding: alternative to colonoscopy in the frail, sick patient, if colonoscopy is unsuccessful or if not acceptable to the patient to diagnose colon cancer, inflammatory bowel disease or diverticular disease; useful in colon cancer screening
Abdominal ultrasound scan	Biliary colic, jaundice, pancreatitis, malignancy
	Gallstones, liver metastases, cholestasis, pancreatic calcification, subphrenic abscess
Abdominal CT	Acute abdomen, suspected pancreatic or renal mass, tumour staging, abdominal aortic aneurysm
	Confirms or excludes metastatic disease and leaking from the aortic aneurysm
MR cholangiopancreatography (MRCP)	Obstructive jaundice, acute and chronic pancreatitis
Pelvic ultrasound scan	Pelvic masses, inflammatory diseases, ectopic pregnancy, polycystic ovary syndrome
	Pelvic structures and abnormalities
	Ascitic fluid
Invasive procedures	
Upper gastrointestinal endoscopy	Dysphagia, dyspepsia, gastrointestinal bleeding, gastric ulcer, malabsorption
	Gastric and/or duodenal biopsies are useful
Lower gastrointestinal endoscopy (colonoscopy)	Rectal bleeding, obscure gastrointestinal bleeding, altered bowel habit, iron deficiency anaemia
	Able to biopsy lesions and remove polyps
Video capsule endoscopy	Obscure gastrointestinal bleeding with bidirectional negative endoscopies, suspected small bowel disease (vascular malformations, inflammatory bowel disease)
Endoscopic retrograde cholangiopancreatography (ERCP)	Obstructive jaundice, acute and chronic pancreatitis
	Mainly therapeutic role
	Stenting strictures and removing stones
Endoscopic ultrasound ± fine-needle aspiration (FNA) or Tru-Cut needle biopsy	Staging of upper gastrointestinal or pancreatobiliary cancer
	Gallstone detection in the biliary tree
	Drainage of pancreatic pseudocysts
Laparoscopy	Suspected appendicitis or perforated viscus, suspected ectopic pregnancy, chronic pelvic pain (e.g. due to endometriosis or pelvic inflammatory disease), suspected ovarian disease (e.g. ruptured ovarian cyst), peritoneal and liver disease
Ultrasound- or CT-guided aspiration cytology and biopsy	Liver metastases, intra-abdominal or retroperitoneal tumours
Liver biopsy	Parenchymal disease of the liver
	Tissue biopsy by percutaneous, transjugular or laparoscopic route
Others	
Pancreatic function tests	Faecal elastase, pancreolauryl test

CT, computed tomography.



A



B



C

Fig. 6.26 Radiography in gastrointestinal disease. [A] Air under the diaphragm on chest x-ray due to a perforated duodenal ulcer. [B] Dilated small bowel due to acute intestinal obstruction. [C] Dilated loop of large bowel due to sigmoid volvulus. WWW.BOOKBAZ.IR

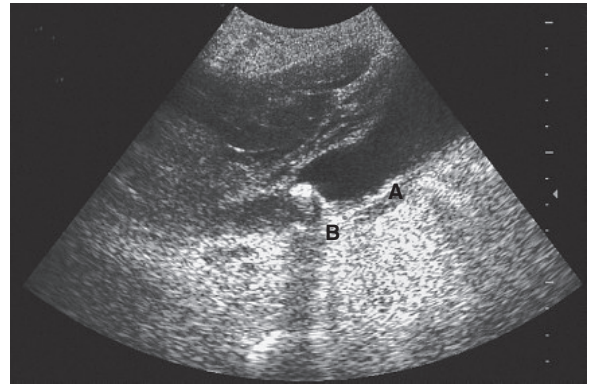
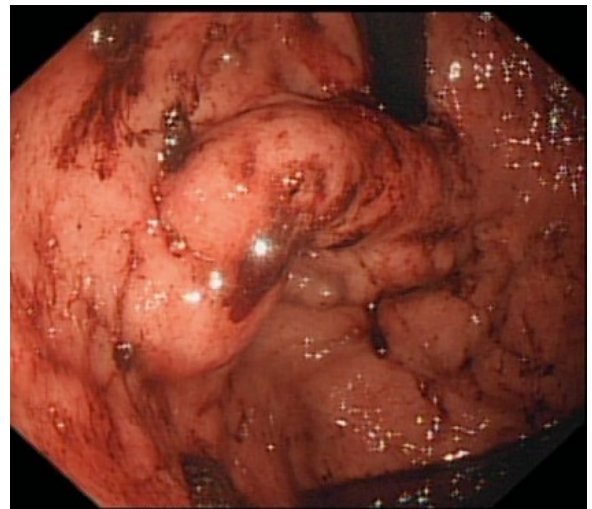


Fig. 6.27 Ultrasound scan of the gallbladder. A, Thick-walled gallbladder containing gallstones. B, Posterior acoustic shadowing.



A



B

Fig. 6.28 Gastrointestinal endoscopy. [A] Gastric ulcer. [B] Gastric varices.

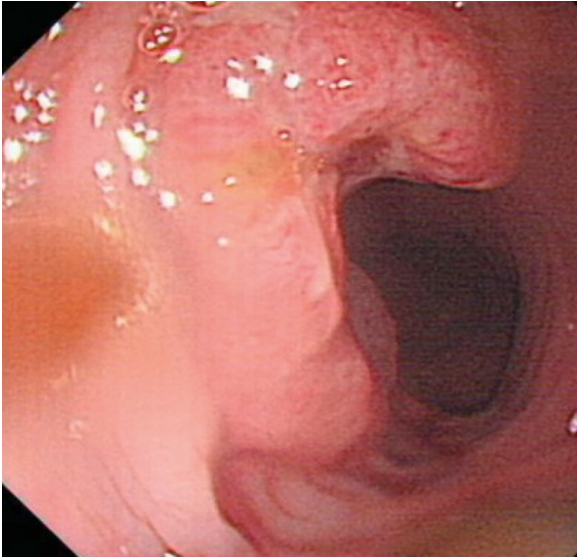


Fig. 6.29 Colonoscopy. Colon cancer.

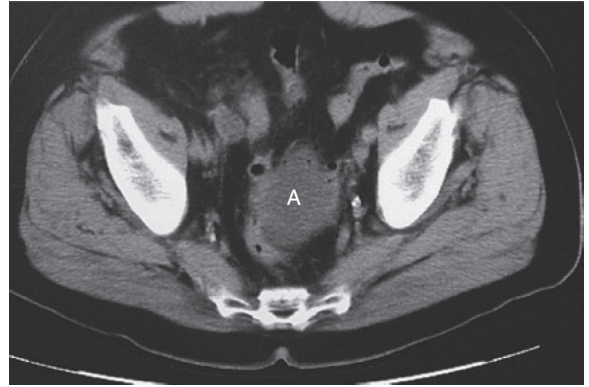


Fig. 6.30 Computed tomogram of the pelvis. A, Diverticular abscess.

OSCE example 1: Abdominal pain and diarrhoea

Mr Reid, 29 years old, presents with a 6-month history of anorexia, 7 kg weight loss, abdominal pains and diarrhoea (liquid stool). He underwent appendicectomy 4 months ago following severe right iliac fossa pain.

Please examine the gastrointestinal system

- Introduce yourself and clean your hands.
- Start with a general inspection: body habitus, signs of dehydration, fever and pallor.
- Inspect the hands: palmar erythema, finger clubbing, leuconychia, koilonychia, nicotine stains and swollen finger or wrist joints.
- Inspect the face: signs of anaemia (pallor, angular stomatitis), swollen lips and aphthous mouth ulcers.
- Inspect the skin: erythema nodosum or pyoderma gangrenosum.
- Inspect the abdomen: laparoscopy/laparotomy scars or skin fistulae.
- Palpate for right iliac fossa tenderness or the presence of a firm, non-tender mass.
- Offer to examine the perianal area for the presence of dusky blue discolouration, oedematous skin tags and the presence of fissures, fistulae or ulcerations.
- Thank the patient and clean your hands.

Summarise your findings

This 29-year-old man with a history of weight loss and diarrhoea appears comfortable at rest but looks thin. He has a recently healed appendicectomy scar, mild periumbilical and left iliac fossa tenderness, and normal bowel sounds.

Suggest a differential diagnosis

The differential diagnosis is Crohn's disease and irritable bowel syndrome.

Suggest initial investigations

Full blood count, C-reactive protein, liver function tests, urea, creatinine and electrolytes, iron studies, vitamin B₁₂ and folate levels, ileocolonoscopy and small bowel magnetic resonance imaging, faecal calprotectin.

OSCE example 2: Jaundice

Mr MacDonald, a 61-year-old retired salesman, presents with increasing tiredness and loss of appetite over 4 months. Two weeks ago, he noticed dark urine and pale stools, and his friends have remarked that his eyes have become yellow. He has drunk a litre of whisky a day for the last 5 years, although recently he has cut down to a bottle of whisky every 3 days.

Please examine this patient's abdomen

- Introduce yourself and clean your hands.
- Unless prompted otherwise, proceed to peripheral examination prior to concentrating on the abdomen.
- Carry out a general inspection: body habitus, evidence of malnutrition, pallor or jaundice, scratch marks on the forearm and back, bruising.
- Examine the hands: palmar erythema, finger clubbing, leuconychia, Dupuytren's contractures.
- Check for a flapping tremor.
- Examine the face: telangiectasias, xanthelasma, bilateral parotid enlargement and jaundice (yellow sclera of the eyes and skin).
- Smell for alcohol or fetor hepaticus.
- Inspect the neck and chest for spider naevi, gynecomastia; look for axillary and chest hair loss.
- Inspect the abdomen for distension, everted umbilicus, caput medusae or scars of recent drain insertion.
- Palpate and percuss the abdomen for hepatomegaly and splenomegaly.
- Percuss for shifting dullness.
- Auscultate for hepatic bruits.
- Look for peripheral oedema.
- Thank the patient and clean your hands.

Summarise your findings

This patient is jaundiced with multiple spider naevi on the chest and abdomen. He has generalised abdominal swelling with shifting dullness and a firm liver edge palpable 2 cm below the costal margin.

Suggest a differential diagnosis

The differential diagnosis is alcoholic cirrhosis, chronic hepatitis and hepatoma.

Suggested initial investigations

Liver function tests, ferritin, viral hepatitis screen, full blood count and prothrombin time, urea, creatinine and electrolytes, alpha-fetoprotein, abdominal ultrasound scan and upper digestive endoscopy (to check for oesophagogastric varices).

Integrated examination sequence for the gastrointestinal system

- Position the patient: supine and comfortable on the examination couch. Expose the abdomen from the xiphisternum to the pubic symphysis.
- Inspection: start with general observation, then inspect the skin, face, neck and chest, and finally the abdomen.
- Palpation:
 - Begin with light, superficial palpation away from any site of pain, then repeat with deeper palpation.
 - Describe any mass and decide whether there is an enlarged abdominal organ.
- Palpation for hepatomegaly:
 - Ask the patient to breathe in deeply through the mouth and feel for the descent of the liver edge on inspiration.
 - Move your hand progressively up the abdomen, between each breath, until you reach the costal margin or detect the liver edge.
- Percussion to confirm hepatomegaly:
 - Ask the patient to hold their breath in full expiration.
 - Percuss for liver dullness and measure the distance in centimetres below the costal margin.
- Palpation and percussion for splenomegaly:
 - Start with your hand over the umbilicus, moving diagonally up and left to feel for the splenic edge as it descends and moves towards the midline on inspiration.
- Check for ascites (shifting dullness):
 - Percuss from the midline out to the flanks for dullness.
 - Keep your finger on the site of dullness in the flank; ask the patient to turn on to their opposite side and then percuss again. If the area of dullness is now resonant, shifting dullness is present.
- Check for a fluid thrill:
 - Place the palm of your left hand flat against the left side of the patient's abdomen and flick a finger of your right hand against the right side of the abdomen. If you still feel a ripple against your left hand, a fluid thrill is present.
- Auscultation:
 - Listen to the right of the umbilicus for bowel sounds, above the umbilicus over the aorta for arterial bruits, lateral to the umbilicus for bruits from renal artery stenosis, and over the liver for hepatic bruits.
- Check for peripheral oedema.
- Consider a rectal examination (always with a chaperone).